

State Labor-Market Data

Consolidated Phase-1 Validation Report

Point-in-time validation of three state-level series for forecasting, lead-lag analysis and backtesting

Continued claims • U3 unemployment • Nonfarm payrolls — 50 states + DC, FRED/ALFRED

OVERALL GATE: ALL THREE DATASETS CLEARED with documented caveats

	Checks	PASS	WARN	FAIL
All datasets	111	75	36	0
Continued claims	43	23	20	0
U3 unemployment	34	26	8	0
Nonfarm payrolls	34	26	8	0

Validated at 2026-07-08 17:30:01

Source FRED / ALFRED archival API (self-contained notebooks; all data fetched from the API)

Suites claims_validation.ipynb • ur_validation.ipynb • nfp_validation.ipynb

Executive summary

This report consolidates the Phase-1 validation of three state-level labor-market series - continued claims, the U3 unemployment rate, and nonfarm payrolls - each validated by its own 43/34/34-check suite against its point-in-time (ALFRED) archive. Across **111 checks** the result is **75 PASS, 36 WARN, 0 FAIL**: all three datasets are **cleared for use** subject to documented, per-dataset handling rules. Every WARN carries a concrete rule; none is an open defect.

The unifying finding: one method, three revision regimes

The three series are validated identically but behave very differently, and the single most important cross-cutting result is how much each is **revised after first release** - which decides how strictly point-in-time (PIT) discipline must be enforced in a backtest:

- **Continued claims - PIT optional.** Only ~3% of weeks are ever revised; current data $\approx$ first release. Vintage data is a nicety, not a necessity. Claims are also the timeliest (weekly, ~2-week lag).
- **U3 unemployment - PIT recommended.** ~86% of months change, but the moves are small and quantized to 0.1pp; benchmark years and COVID exceed 1pp. Vintage features are advisable.
- **Nonfarm payrolls - PIT mandatory.** ~99% of months are revised and the annual QCEW benchmark moves estimates materially; a naive current-data backtest carries a median 0.6% look-ahead gap per state-month. Vintage features are required.

Two mechanics are shared and were verified in all suites: (1) the **two-clock discipline** - every value carries an observation timestamp (the period measured) and an information timestamp (when it became public); features must join on the latter. (2) U3 and NFP are published in the **same monthly State Employment and Unemployment report** (~50-day lag, Friday), so they align directly with each other, while claims run on a separate weekly calendar.

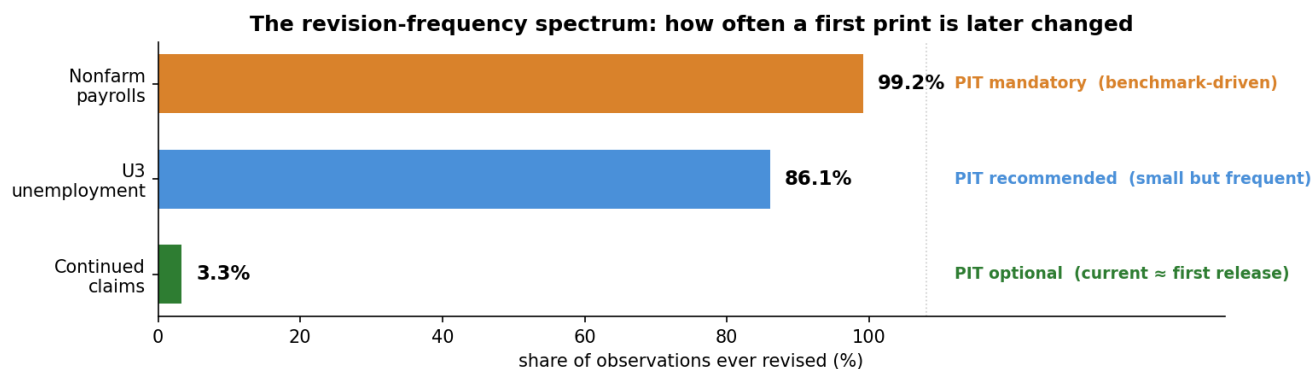
Combined gate decisions

G1 - Scope	Each dataset is cleared for PIT backtesting from its ALFRED ramp-in (claims 2010-11, U3 and NFP 2008); full current history is cleared for climatology/trend estimation only. Pre-vintage data is never point-in-time.
G2 - Transforms	Match the transform to the data type: claims and NFP are counts $\rightarrow$ log/growth; U3 is a bounded rate $\rightarrow$ percentage-point changes. NFP levels trend ($I(1)$); U3 and claims are cyclical. All SA series verified free of residual seasonality; regime-flag 2020 everywhere.
G3 - Point-in-time & aggregation	Enforce PIT in proportion to the revision regime above. Reconcile to national by the right operator: sum for counts (claims, NFP), labor-force-weighted mean for the U3 rate. Apply annual benchmarks on their release date, never spread backward.

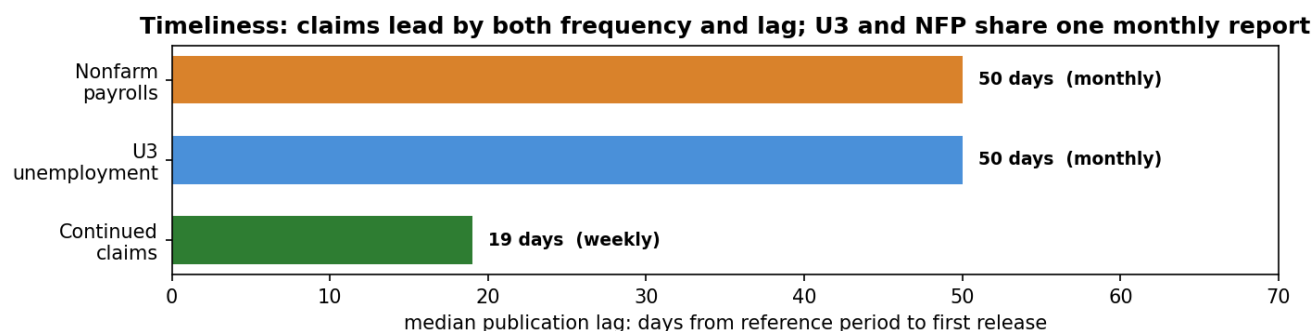
Cross-dataset comparison

The three series share a validation method but occupy three distinct points in data-space. This section is the payoff of validating them together.

	Continued claims	U3 unemployment	Nonfarm payrolls
Data type	level (weekly count)	rate (percent, bounded)	level (monthly count)
Frequency	weekly (Sat week-end)	monthly (week of 12th)	monthly (pay period, 12th)
Seasonal adj.	NSA	SA	SA
Revised?	~3% of obs (rare)	~86% (small, 0.1pp-quantized)	~99% (benchmark-driven)
Level behaviour	cyclical	cyclical, bounded	trending (I(1))
Feature transform	log / % change	percentage-point change	log / % growth
Reconcile to national	sum (PR/VI gap)	labor-force-weighted mean	sum (= exact)
Publication lag	~13-19 days	~50 days	~50 days
PIT requirement	optional	recommended	mandatory
Backtest from	2010-11	2008-08	2008-08
Checks (P/W/F)	23/20/0	26/8/0	26/8/0



The revision-frequency spectrum. Share of observations ever changed after first release: claims 3.3%, U3 86.1%, NFP 99.2%. This single axis sets how strictly point-in-time discipline must be enforced - from optional (claims) to mandatory (NFP).



Timeliness. Continued claims lead on both counts - weekly and a ~2-week lag - while U3 and NFP share a single monthly report released ~50 days after the reference month.

Continued claims

{ABBR}CCLAIMS (weekly, NSA) — the timely, barely-revised weekly series — 23 PASS / 20 WARN / 0 FAIL

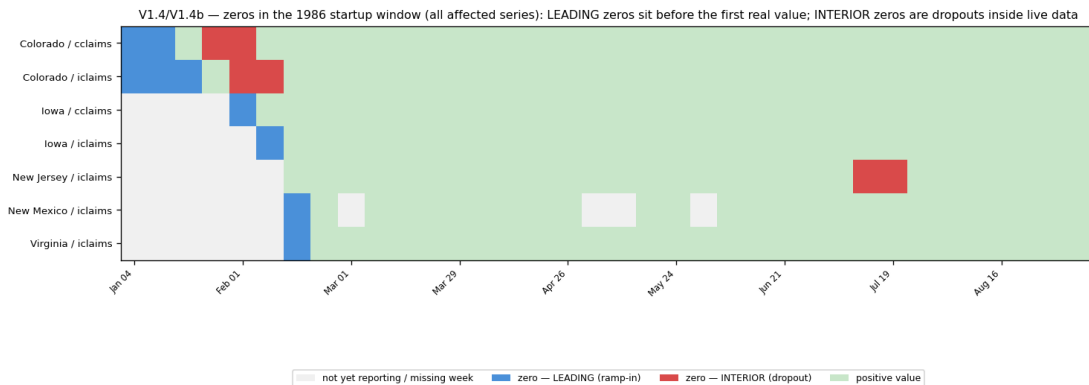
Headline findings

- **Revisions are rare and concentrated.** Only 3.3% of weeks (1,402) ever change; the visible outliers are isolated first-print data errors (e.g. TX 2024-06 dropped a digit) concentrated in a few chronic-reviser states and 2020-22, not a systematic wedge.
- **A 2020 capture-regime switch.** FRED changed how it archives state claims in July 2020 - it stores the *revised* print before and the *advance* (first) print after - so backtests must apply per-era availability lags.
- **Data holds are the real hazard.** California's late-2020 EDD processing pause froze first-release prints (below), which a live model would have consumed as real signal.
- **No look-ahead.** The PIT panel passes a hard guard on all 41,667 rows; minimum availability lag 13 days, always after the public release.

Gate decisions

- PIT is **optional** (current $\approx$ first release) but vintage data still needed for the 2020 print-type switch and the frozen-print episodes.
- Work in log/percent changes; reconcile to national by summing states (expect a stable $\sim 1.4\%$ PR/VI gap).
- Backtest window opens 2010-11; flag the enumerated revision/stale-run exceptions.

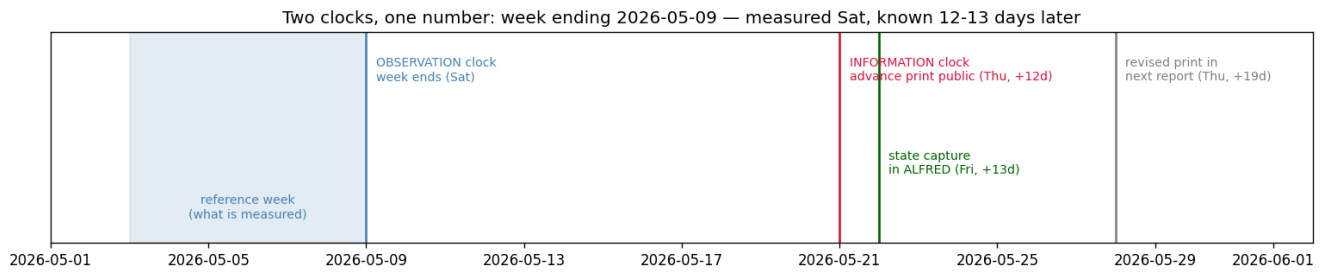
Full diagnostic gallery (21 figures)



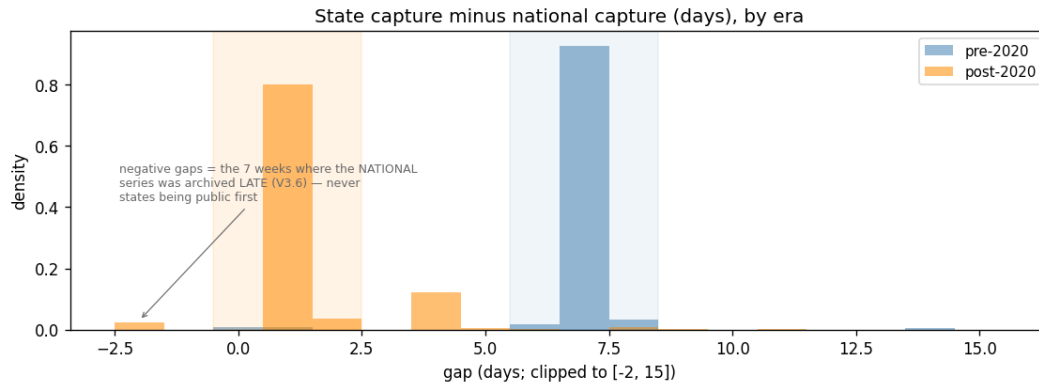
claims_fig_01. V1.4/V1.4b — zeros in the 1986 startup window (all affected series):



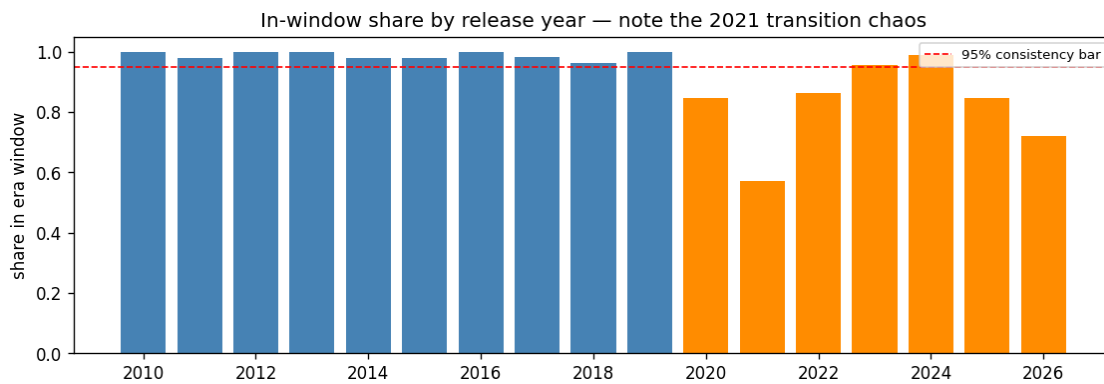
claims_fig_02. Data holes by state — OBSERVATION clock: reference weeks (obs_date) missing from current history



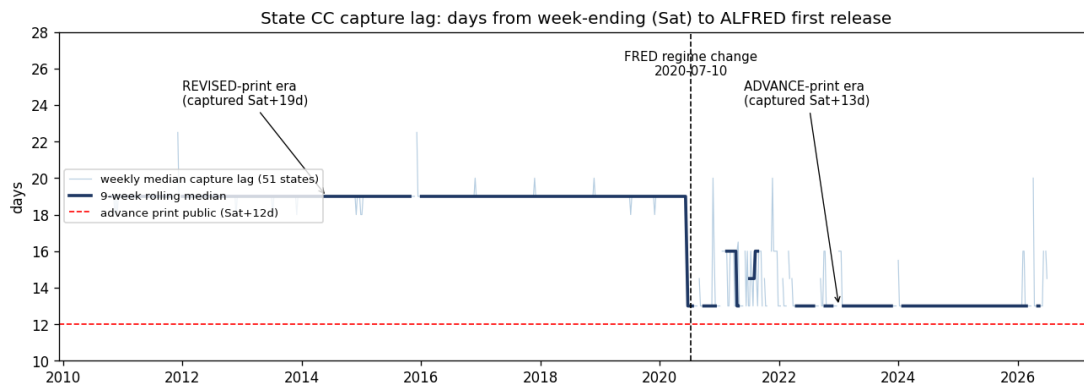
claims_fig_03. Two clocks, one number: week ending the reference week — measured Sat, known 12-13 days later



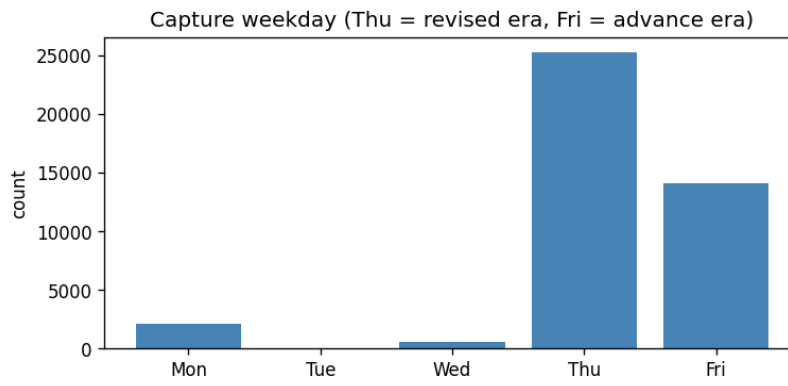
claims_fig_04. State capture minus national capture (days), by era



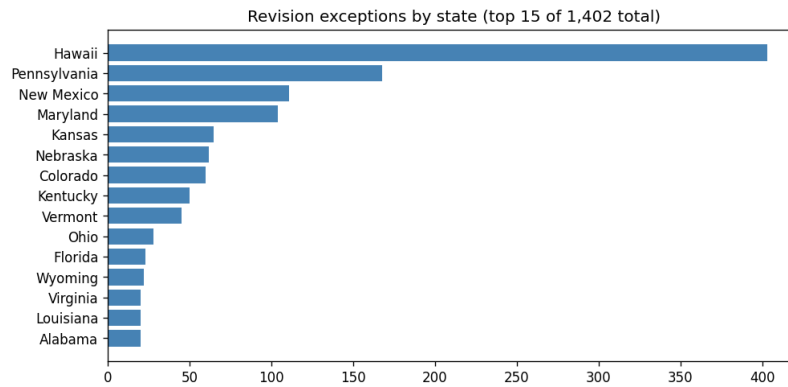
claims_fig_05. In-window share by release year — note the 2021 transition chaos



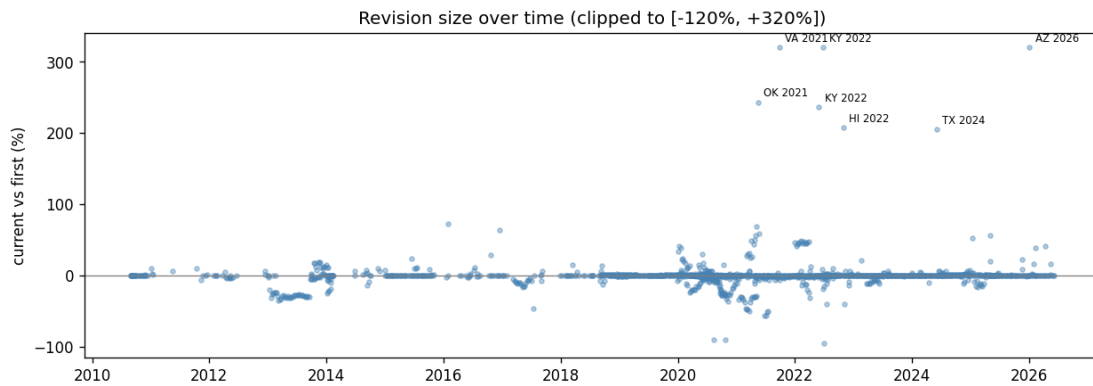
claims_fig_06. State CC capture lag: days from week-ending (Sat) to ALFRED first release



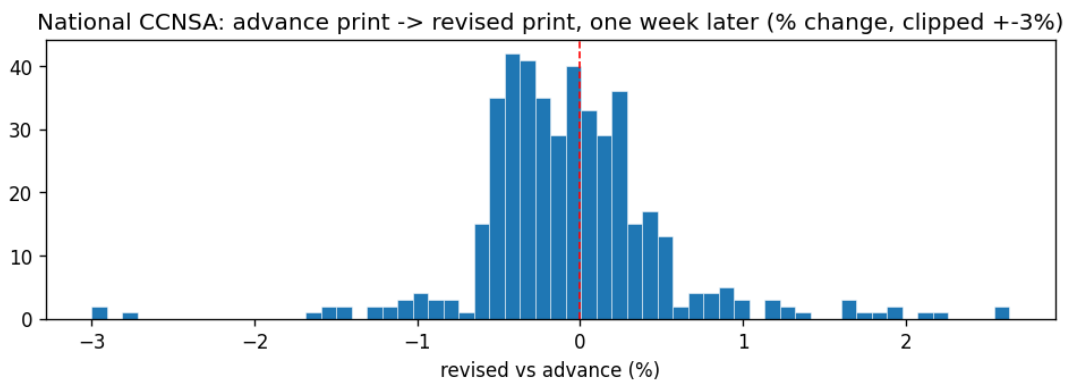
claims_fig_07. Capture weekday (Thu = revised era, Fri = advance era)



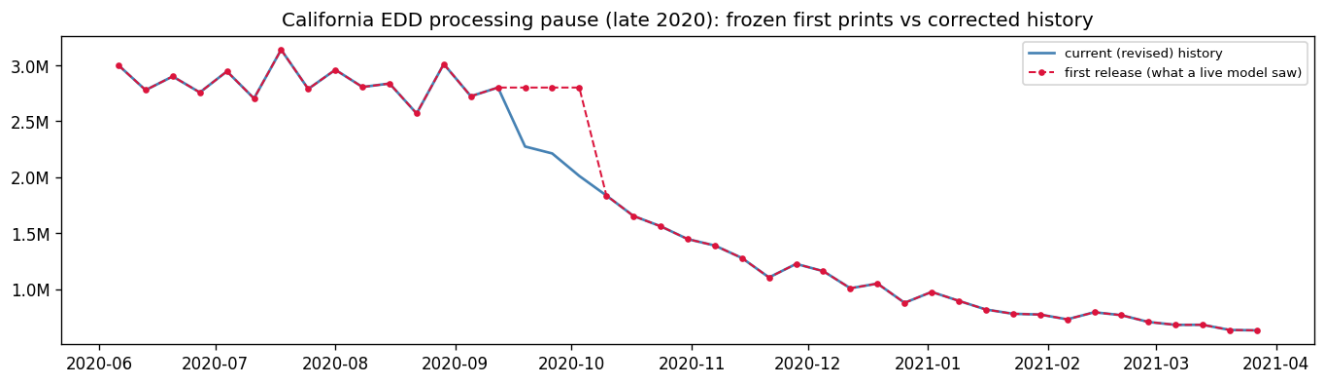
claims_fig_08. Revision exceptions by state (top 15 of 1,402 total)



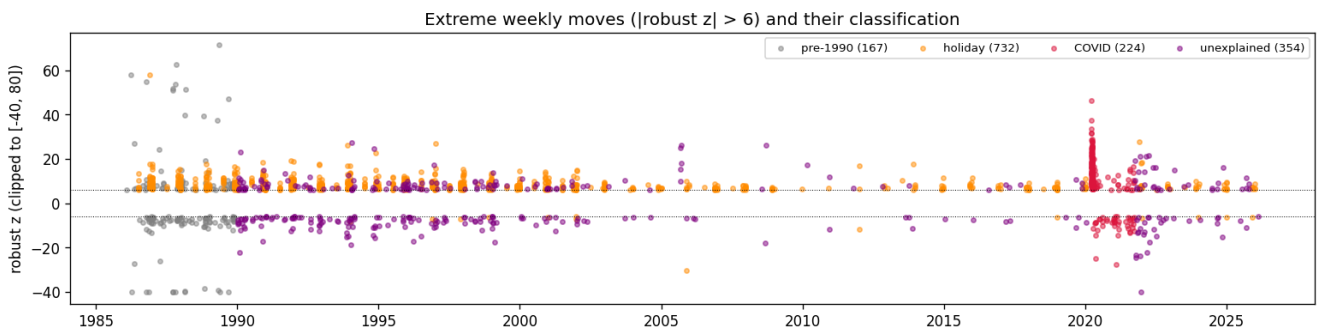
claims_fig_09. Revision size over time (clipped to [-120%, +320%])



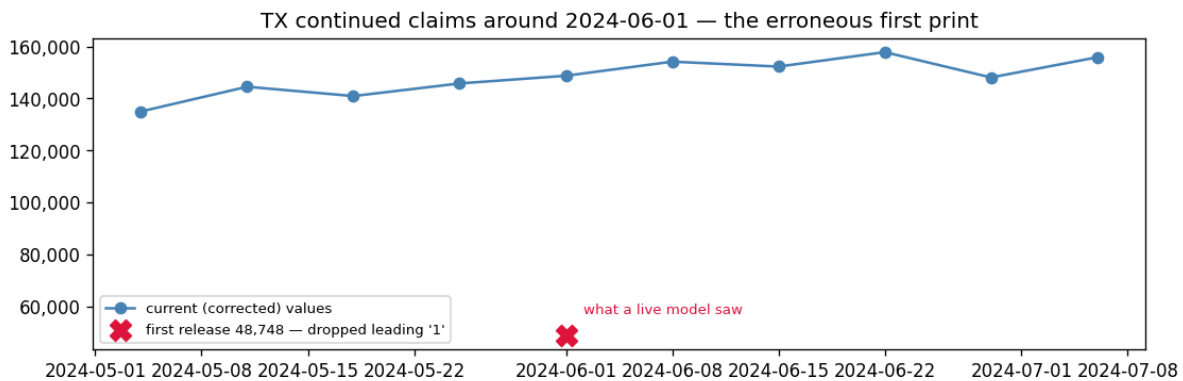
claims_fig_10. National CCNSA: advance print -> revised print, one week later (% change, clipped +3%)



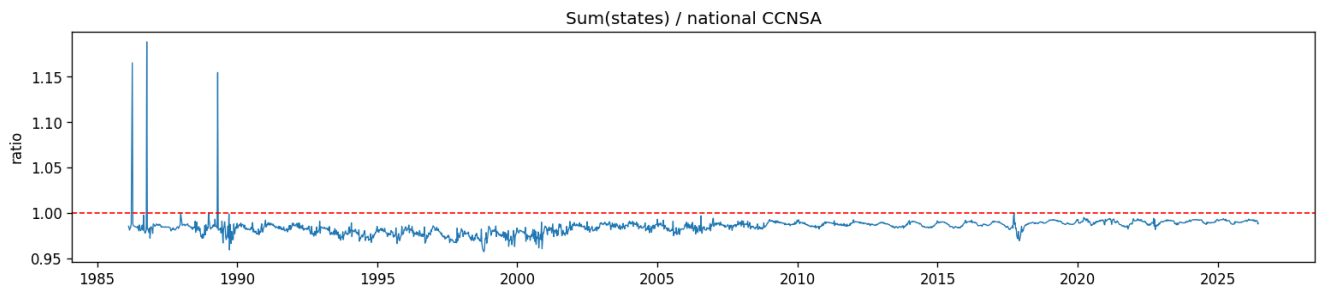
claims_fig_11. California EDD processing pause (late 2020): frozen first prints vs corrected history



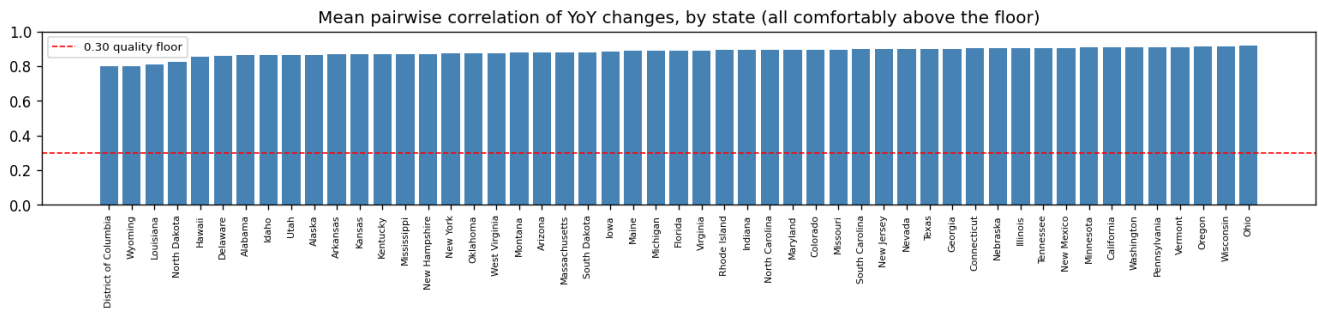
claims_fig_12. Extreme weekly moves ($|robust\ z| > 6$) and their classification



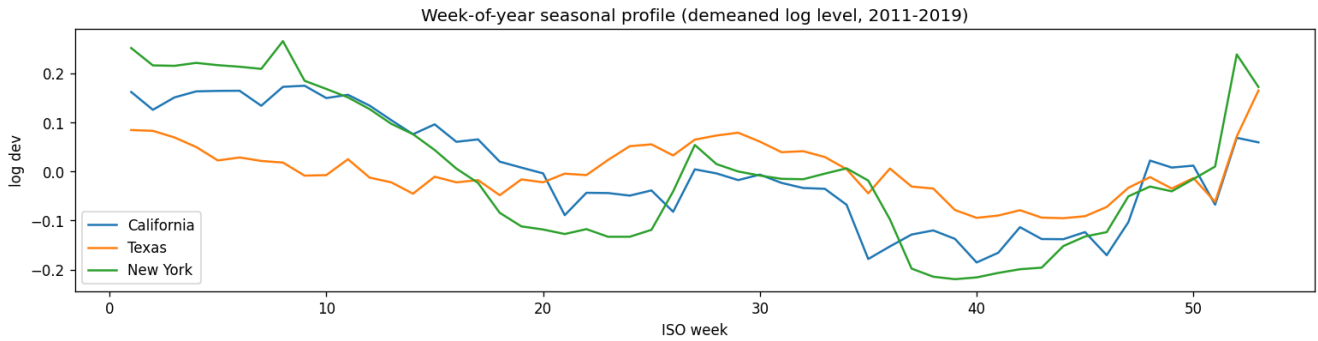
claims_fig_13. TX continued claims around 2024-06-01 — the erroneous first print



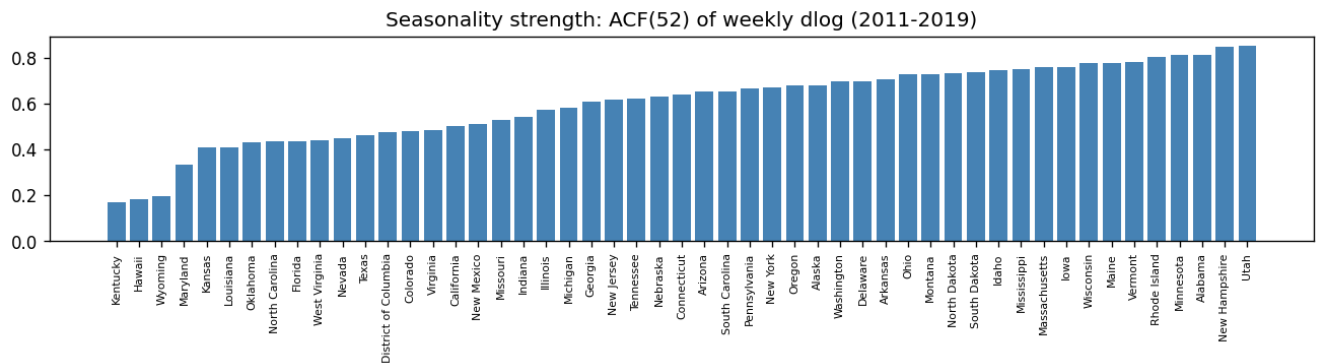
claims_fig_14. Sum(states) / national CCNSA



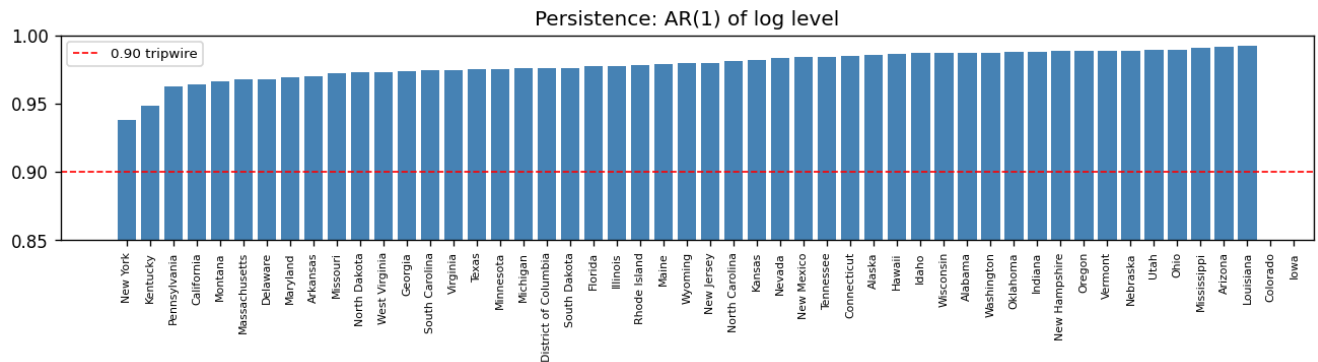
claims_fig_15. Mean pairwise correlation of YoY changes, by state (all comfortably above the floor)



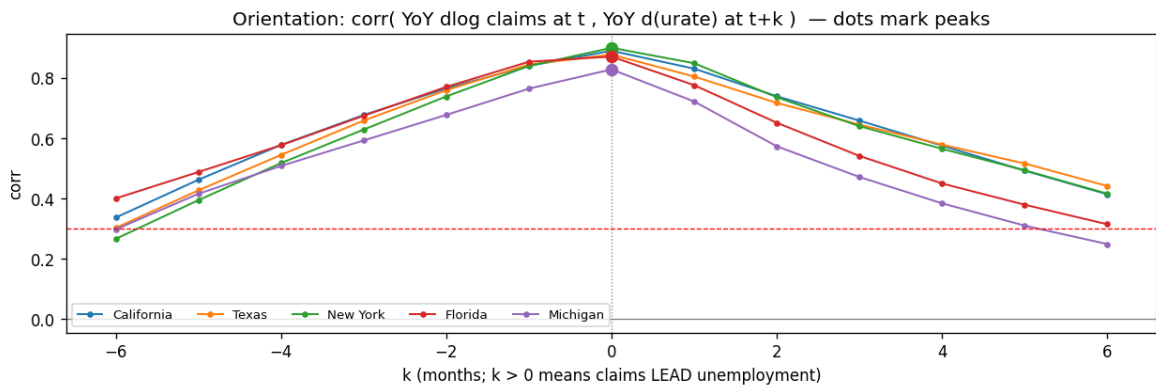
claims_fig_16. Week-of-year seasonal profile (demeaned log level, 2011-2019)



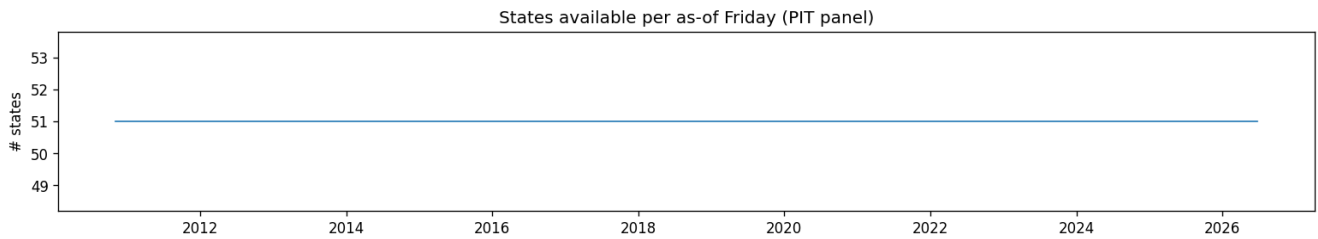
claims_fig_17. Seasonality strength: ACF(52) of weekly dlog (2011-2019)



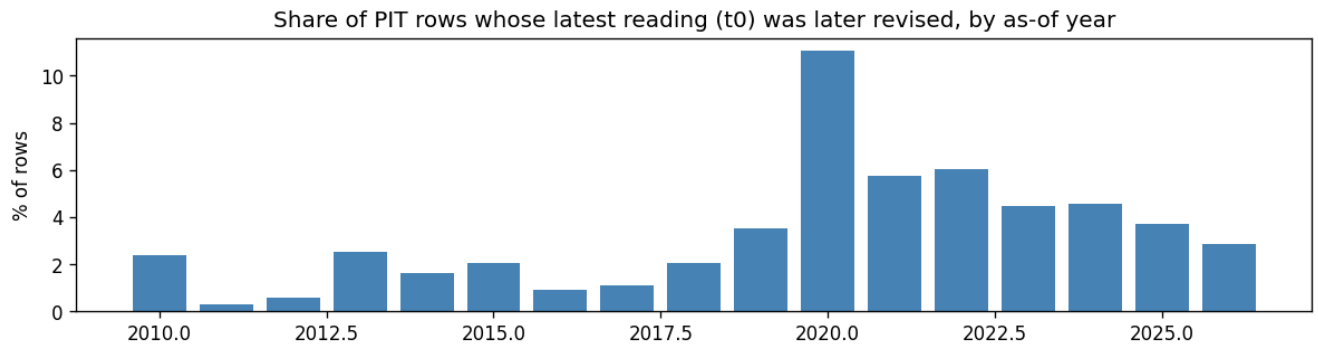
claims_fig_18. Persistence: AR(1) of log level



claims_fig_19. Orientation: corr(YoY dlog claims at t , YoY d(urate) at t+k) — dots mark peaks



claims_fig_20. States available per as-of Friday (PIT panel)



claims_fig_21. Share of PIT rows whose latest reading (t0) was later revised, by as-of year

U3 unemployment

{ABBR}UR (monthly, SA rate) — the bounded, moderately-revised rate — 26 PASS / 8 WARN / 0 FAIL

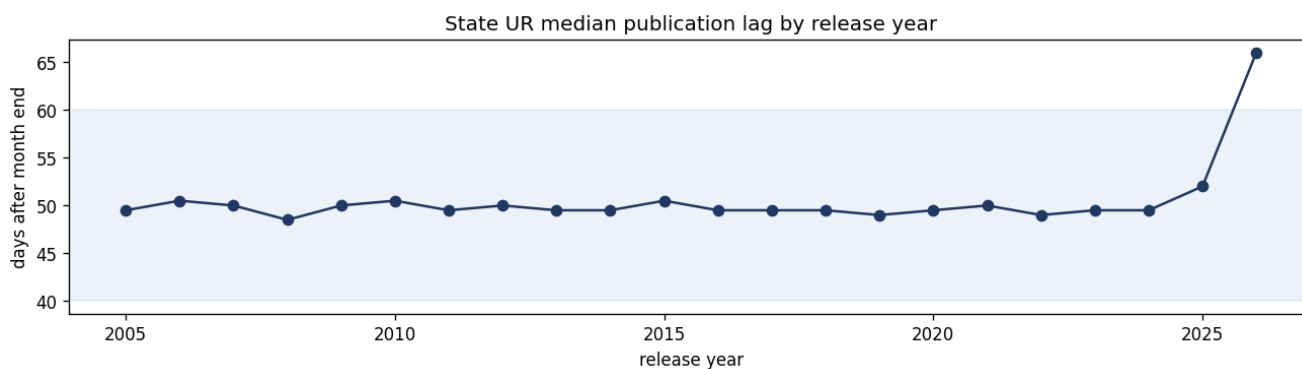
Headline findings

- **A bounded rate, not a count.** Values sit in [1.7%, 30.5%], quantized to 0.1pp; features are percentage-point changes, and the level is cyclical and mean-reverting (AR1 0.96-0.98) - never trending.
- **Reconciliation needs weighting.** The labor-force-weighted mean of state rates tracks national UNRATE to 0.06pp, while the naive unweighted average is biased 0.32pp low - you cannot average rates.
- **Moderate, quantized revisions.** 86% of months change but most within the 0.1pp rounding; the annual benchmark drives the larger ones (99% of the biggest land in the Jan-Apr window).
- **A real recent data hole.** October 2025 is missing for every state and for national UNRATE - the late-2025 federal shutdown delayed the release. Flagged, not interpolated.

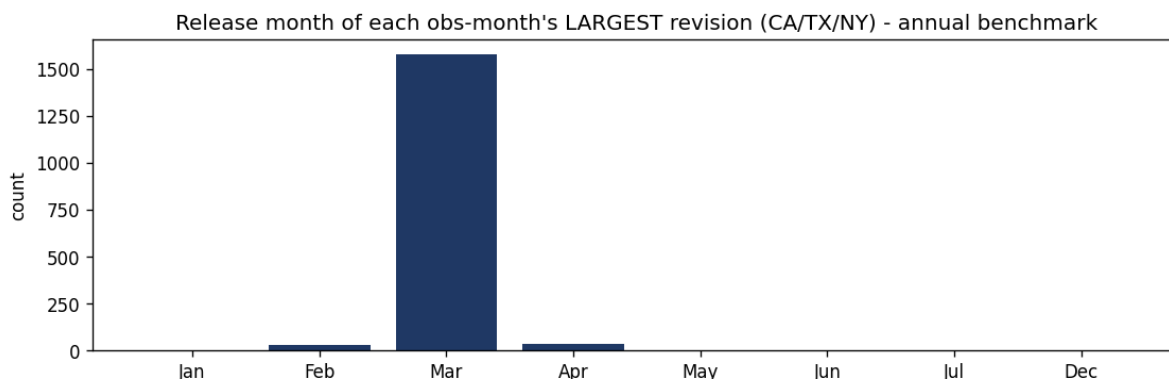
Gate decisions

- PIT is **recommended**: revisions are small but frequent; use vintage features, current-data gap is a median 0.20pp (p90 0.70pp).
- Model percentage-point changes (bounded, cyclical); SA already removes seasonality (residual R^2 0.004).
- Reconcile by labor-force-weighted mean; backtest window opens 2008-08; handle the Oct-2025 hole as missing.

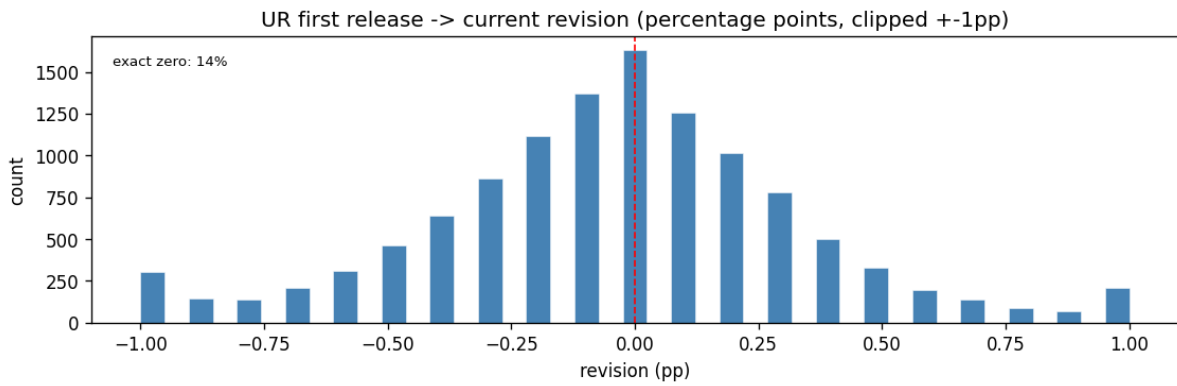
Full diagnostic gallery (18 figures)



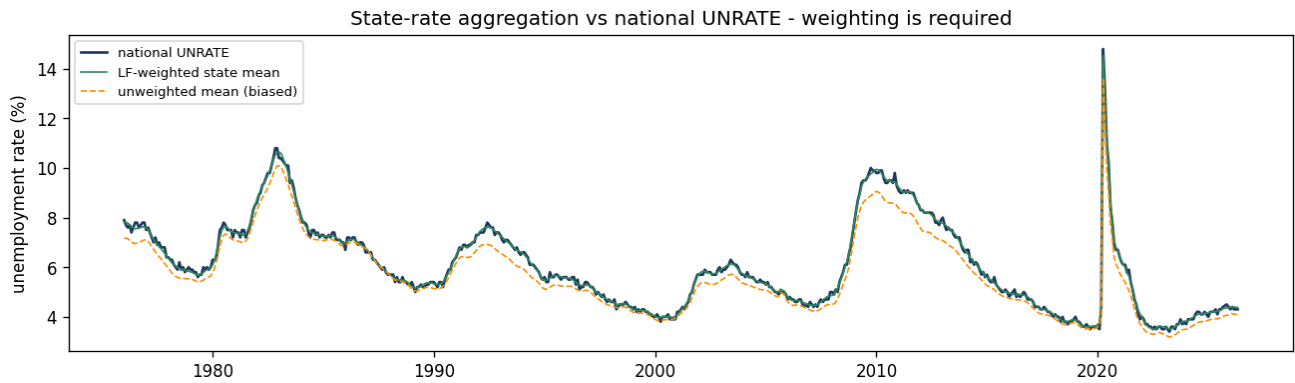
ur_fig_01. State UR median publication lag by release year



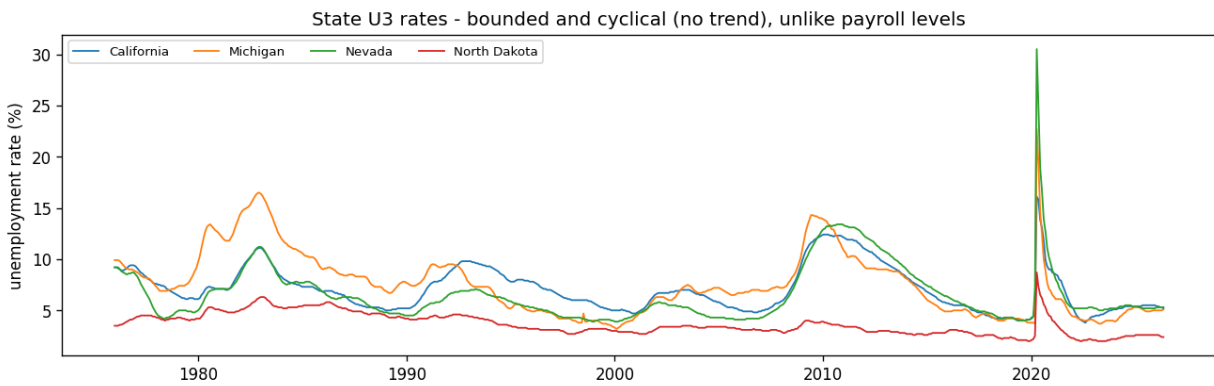
ur_fig_02. Release month of each obs-month's LARGEST revision (CA/TX/NY) - annual benchmark



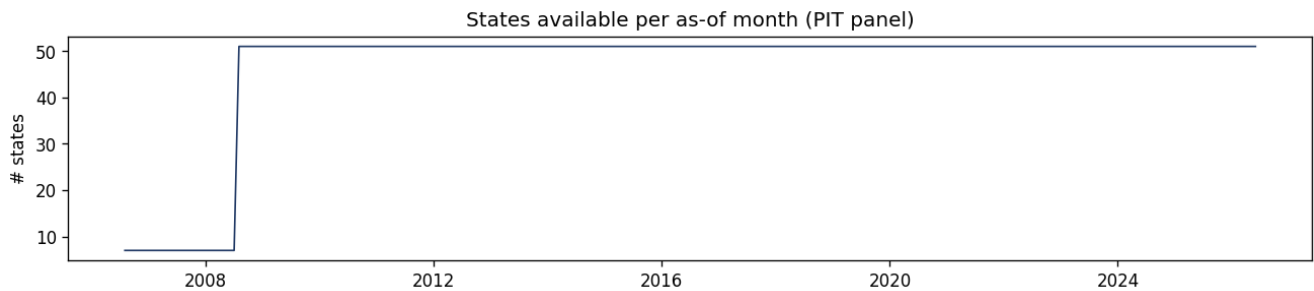
ur_fig_03. UR first release -> current revision (percentage points, clipped +-1pp)



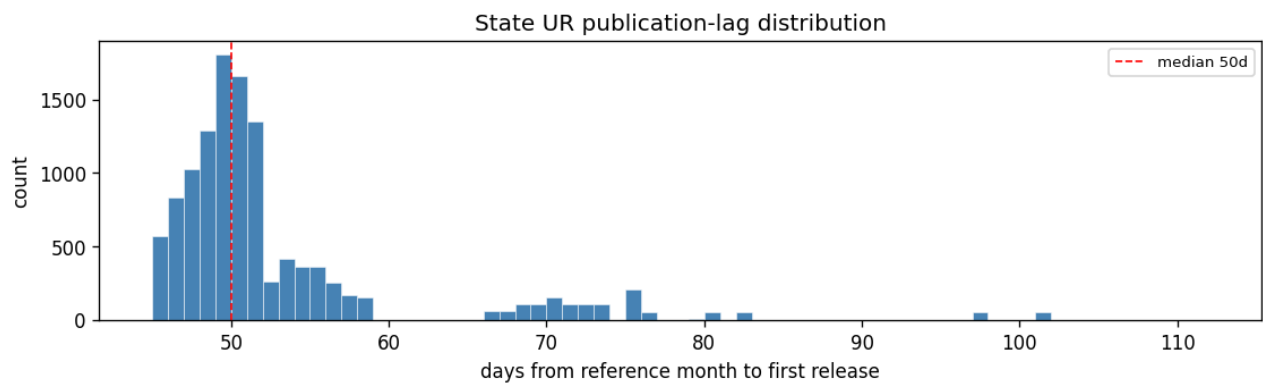
ur_fig_04. State-rate aggregation vs national UNRATE - weighting is required



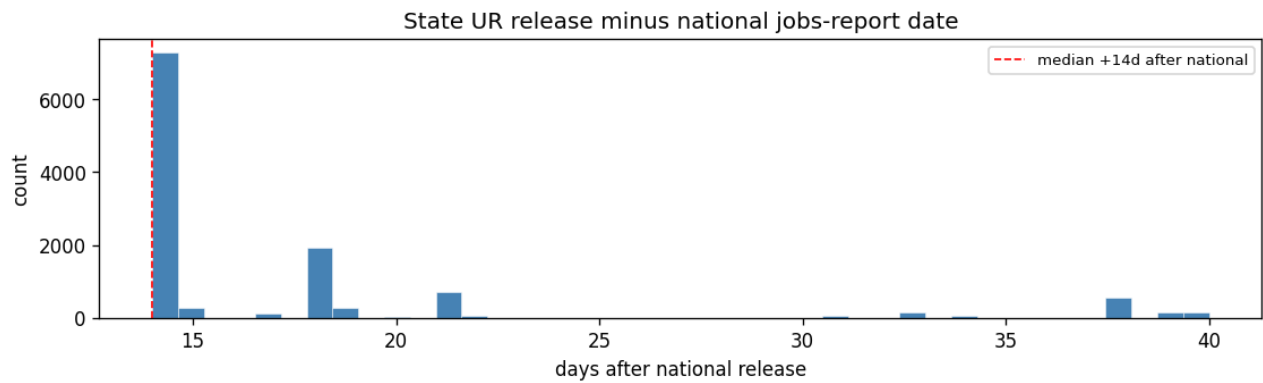
ur_fig_05. State U3 rates - bounded and cyclical (no trend), unlike payroll levels



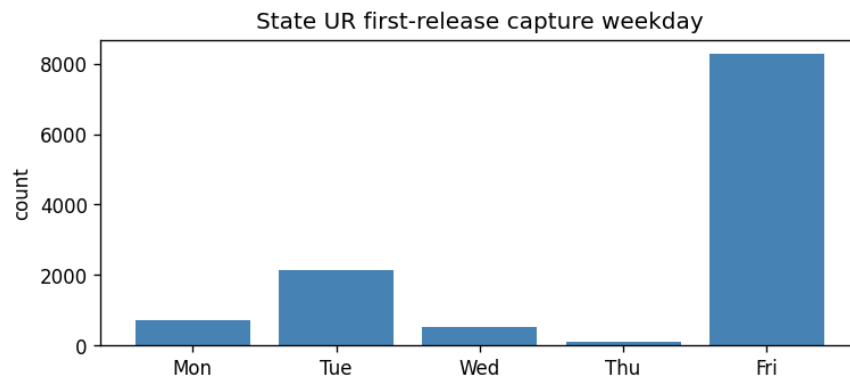
ur_fig_06. States available per as-of month (PIT panel)



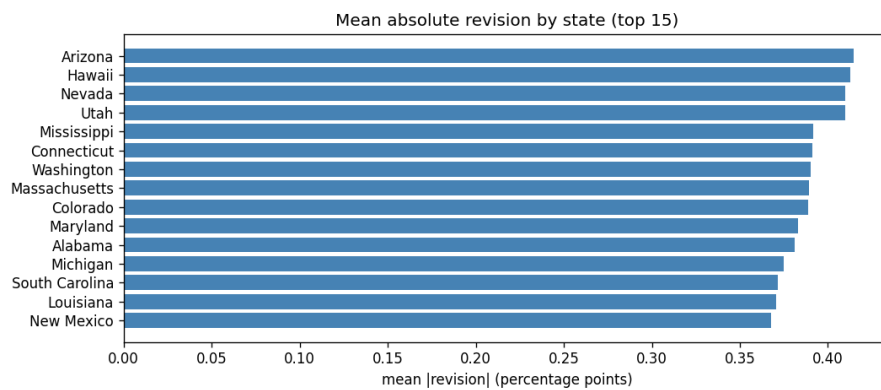
ur_fig_07. State UR publication-lag distribution



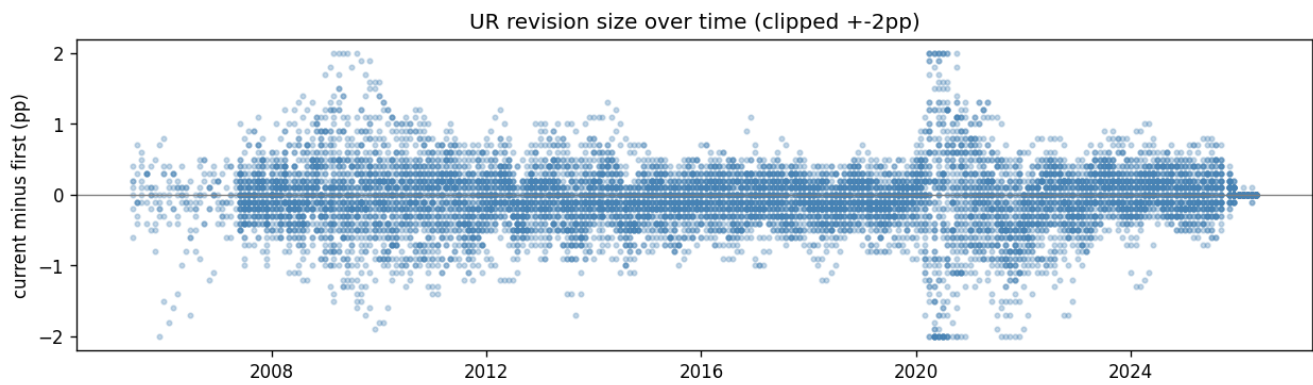
ur_fig_08. State UR release minus national jobs-report date



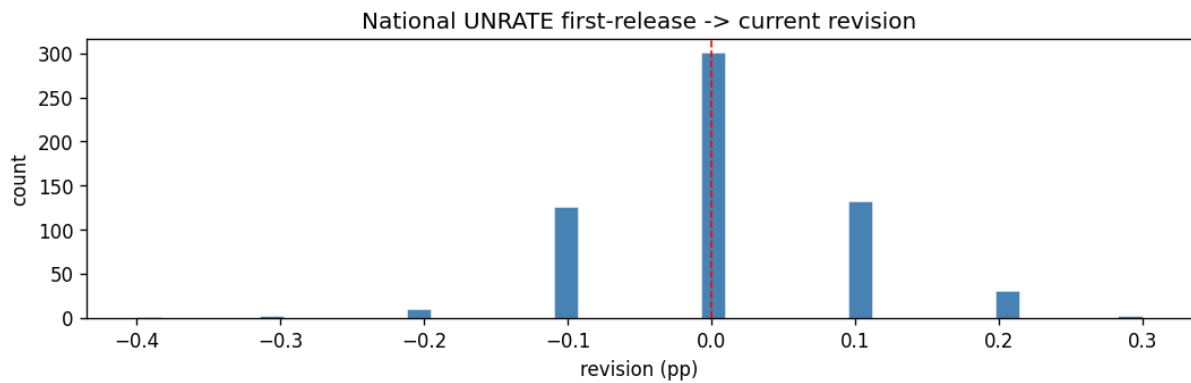
ur_fig_09. State UR first-release capture weekday



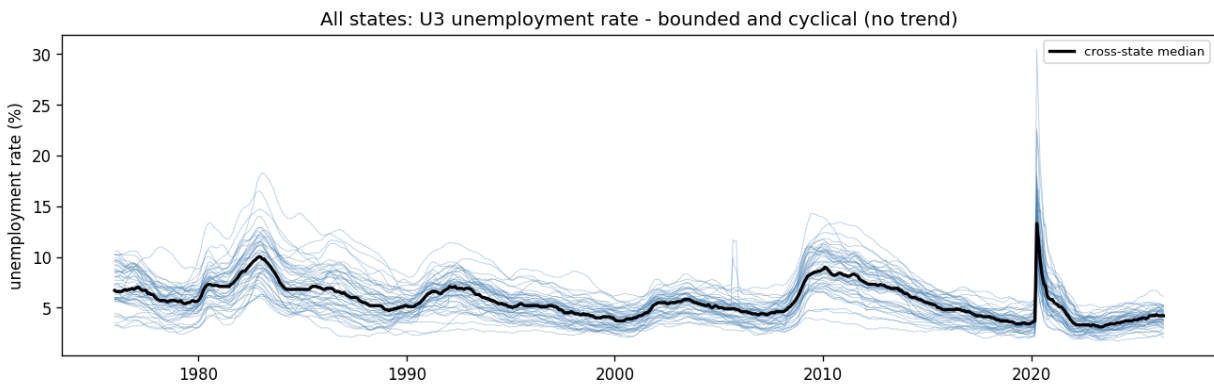
ur_fig_10. Mean absolute revision by state (top 15)



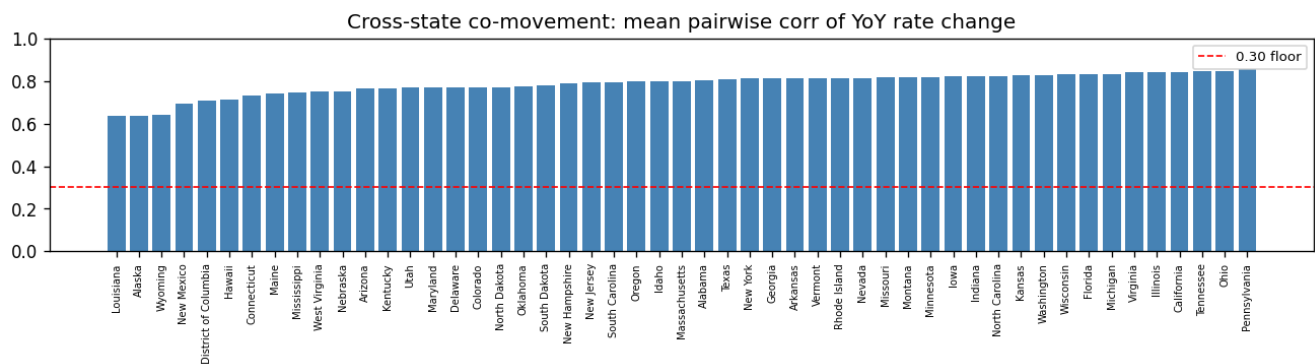
ur_fig_11. UR revision size over time (clipped +2pp)



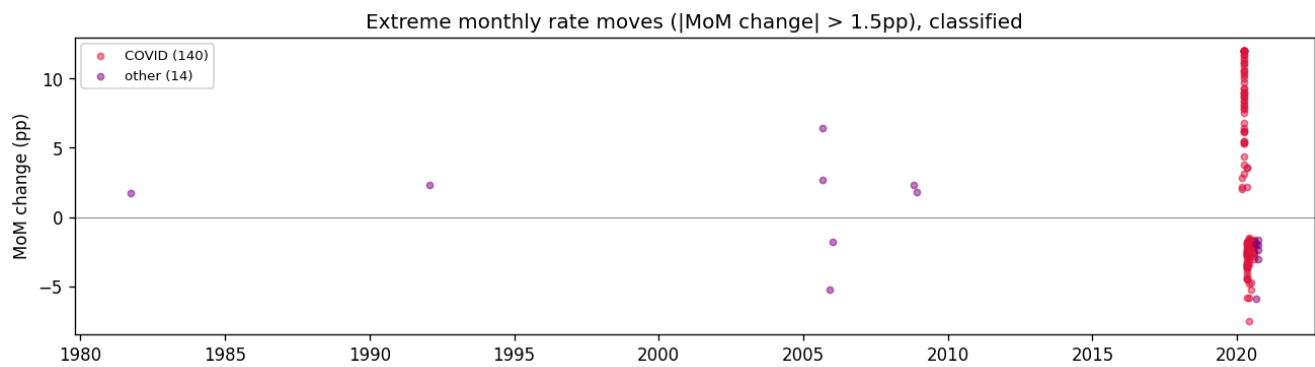
ur_fig_12. National UNRATE first-release -> current revision



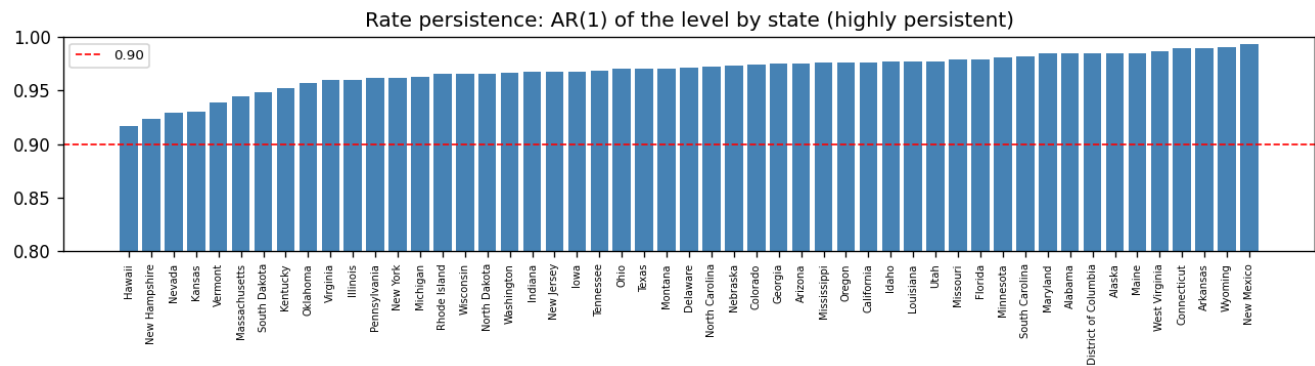
ur_fig_13. All states: U3 unemployment rate - bounded and cyclical (no trend)



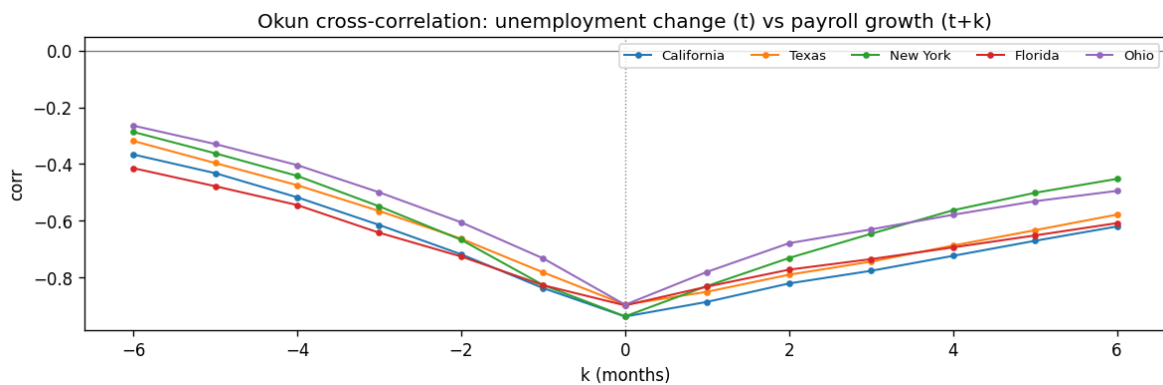
ur_fig_14. Cross-state co-movement: mean pairwise corr of YoY rate change



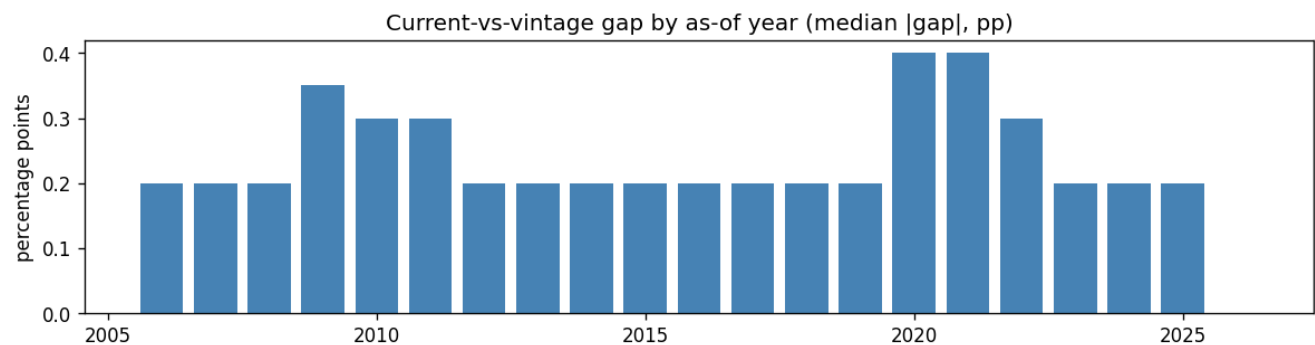
ur_fig_15. Extreme monthly rate moves ($|\text{MoM change}| > 1.5\text{pp}$), classified



ur_fig_16. Rate persistence: AR(1) of the level by state (highly persistent)



ur_fig_17. Okun cross-correlation: unemployment change (t) vs payroll growth (t+k)



ur_fig_18. Current-vs-vintage gap by as-of year (median $|\text{gap}|$, pp)

Nonfarm payrolls

{ABBR}NA (monthly, SA level) — the heavily-revised, trending level — 26 PASS / 8 WARN / 0 FAIL

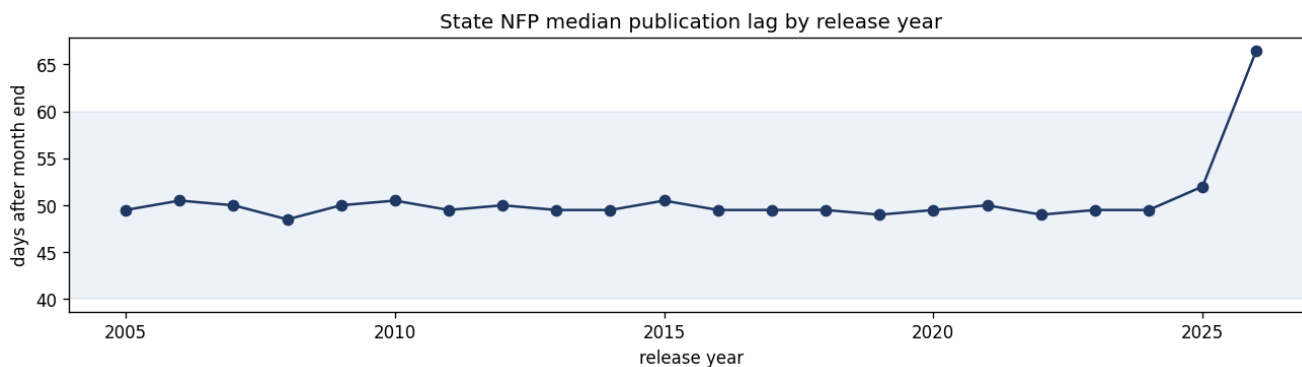
Headline findings

- **Point-in-time is mandatory.** 99.2% of months are revised; current data embeds annual QCEW benchmarks unknown in real time. A naive current-data backtest carries a median 0.6% (p90 1.7%) look-ahead gap per state-month - the central backtesting finding.
- **The benchmark is the story.** 95% of each month's largest revision lands in the winter benchmark window (modal March); the first-release vs current wedge (below) is largest around recessions and 2020.
- **Trending levels, stationary growth.** Log levels are I(1) with drift (non-stationary); model payroll growth. This is the sharpest contrast with the cyclical claims and U3 series.
- **Seasonal adjustment works and states reconcile.** SA month-of-year R^2 is 0.01 vs 0.88 for NSA; the 51 states sum to national PAYEMS at a ratio of 1.0001.

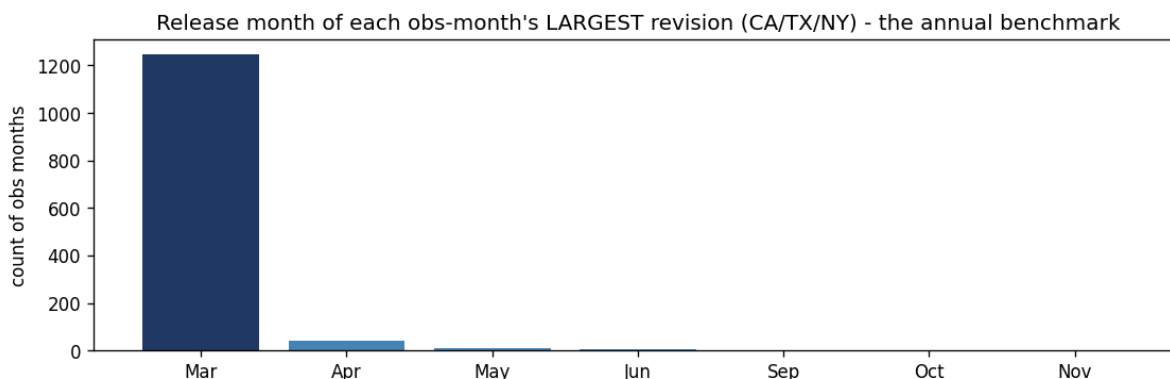
Gate decisions

- PIT is **mandatory**: use first-release / vintage features (nfp_pit_features.csv); never train on current data.
- Model payroll growth (MoM/YoY log-diff) - levels are trending and non-stationary; use SA; regime-flag 2020.
- Reconcile by summing states ($\approx$ exact); apply benchmarks on release date; backtest window opens 2008-08.

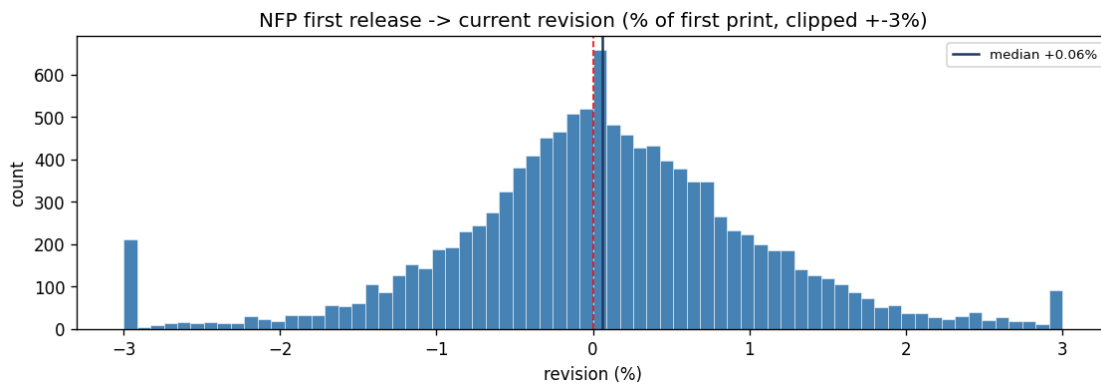
Full diagnostic gallery (19 figures)



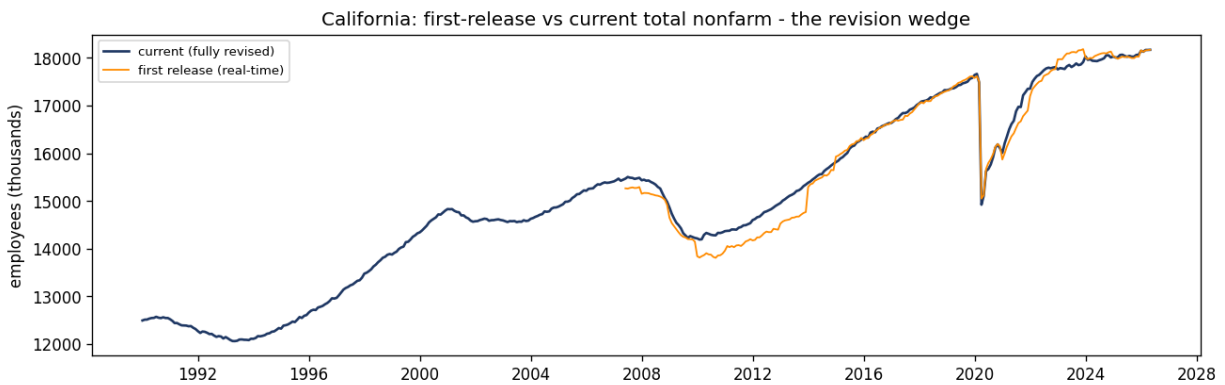
nfp_fig_01. State NFP median publication lag by release year



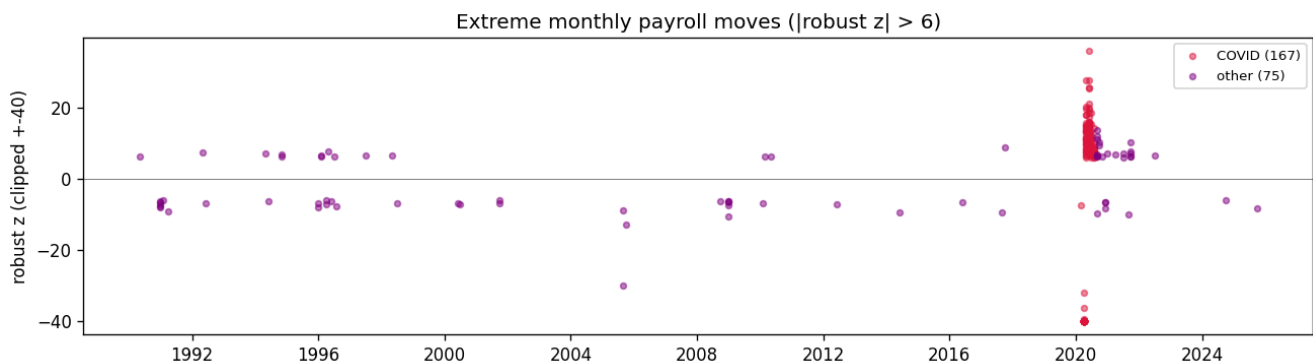
nfp_fig_02. Release month of each obs-month's LARGEST revision (CA/TX/NY) - the annual benchmark



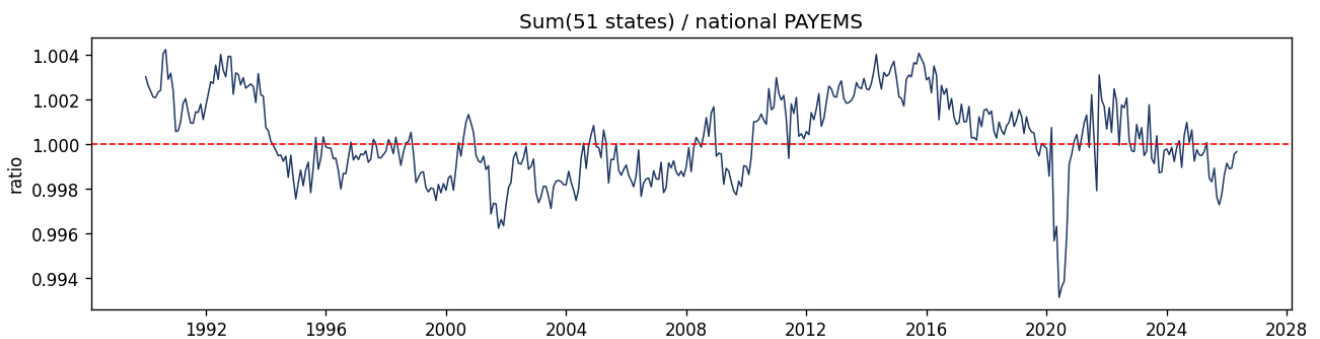
nfp_fig_03. NFP first release -> current revision (% of first print, clipped +-3%)



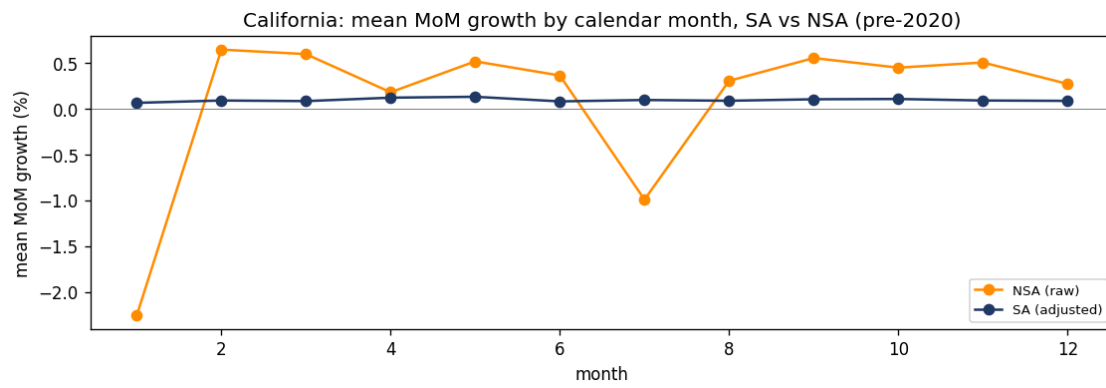
nfp_fig_04. California: first-release vs current total nonfarm - the revision wedge



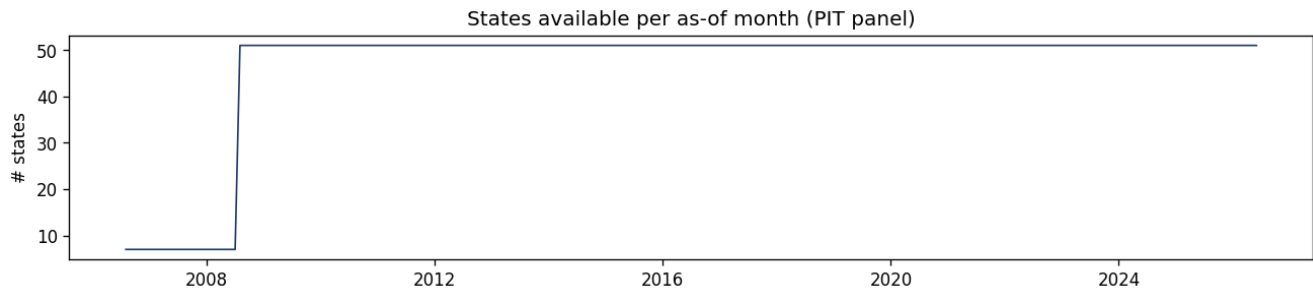
nfp_fig_05. Extreme monthly payroll moves ($|robust\ z| > 6$)



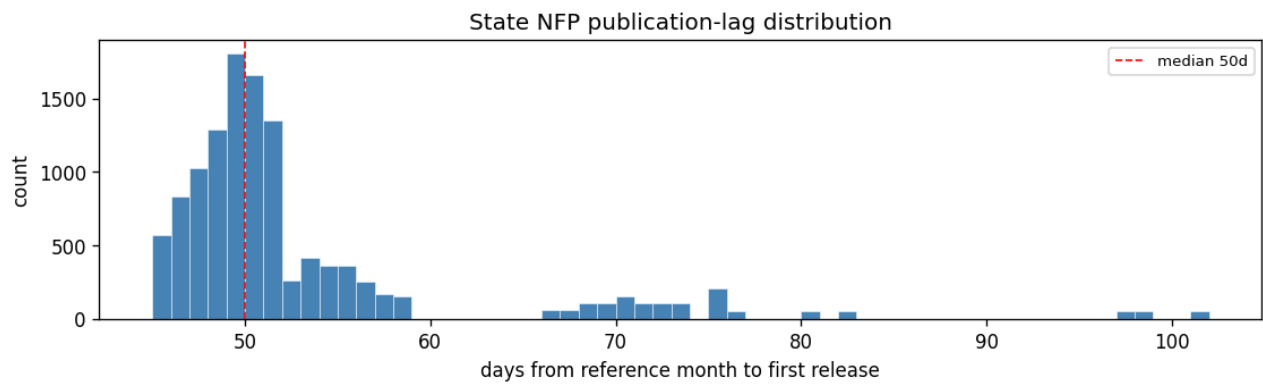
nfp_fig_06. Sum(51 states) / national PAYEMS



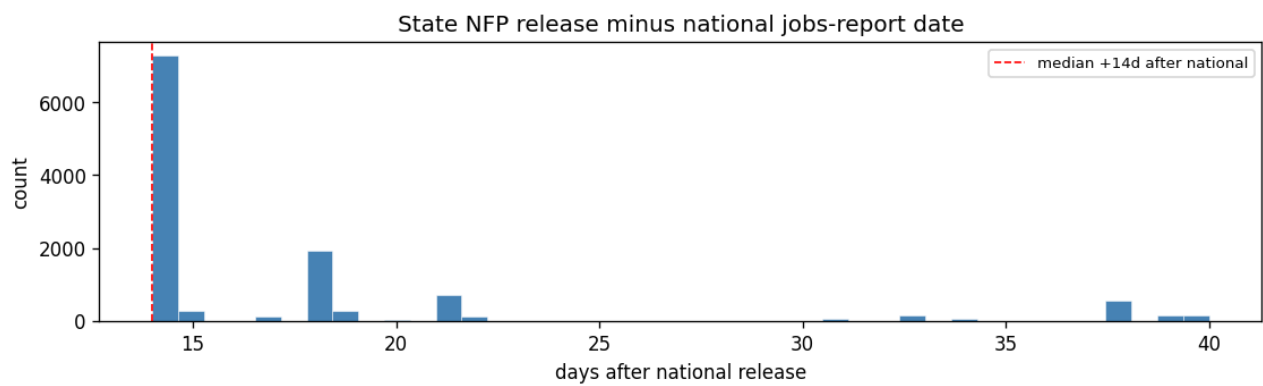
nfp_fig_07. California: mean MoM growth by calendar month, SA vs NSA (pre-2020)



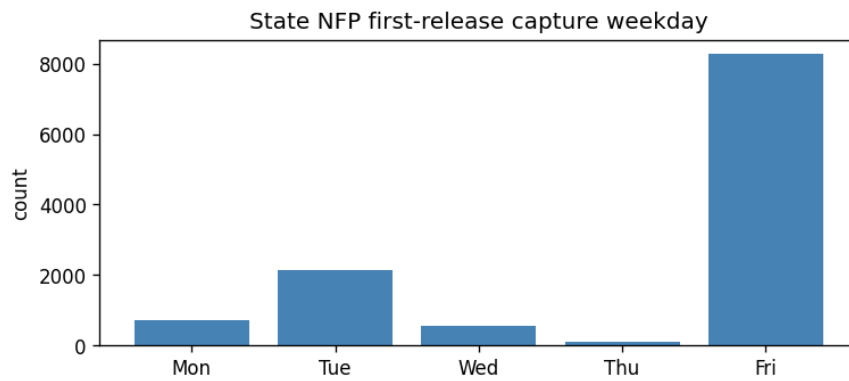
nfp_fig_08. States available per as-of month (PIT panel)



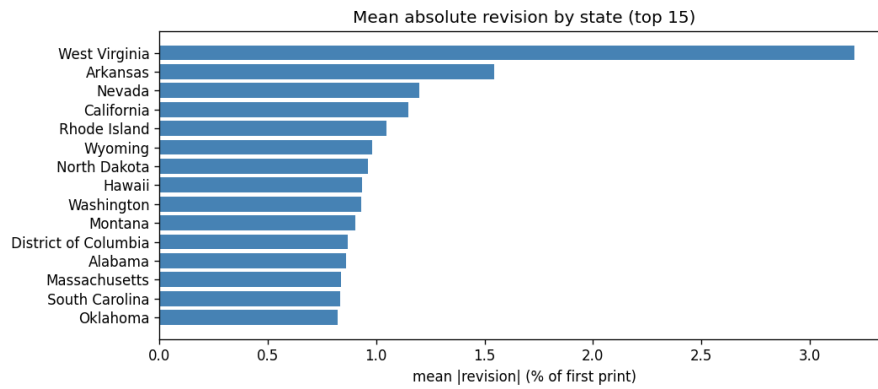
nfp_fig_09. State NFP publication-lag distribution



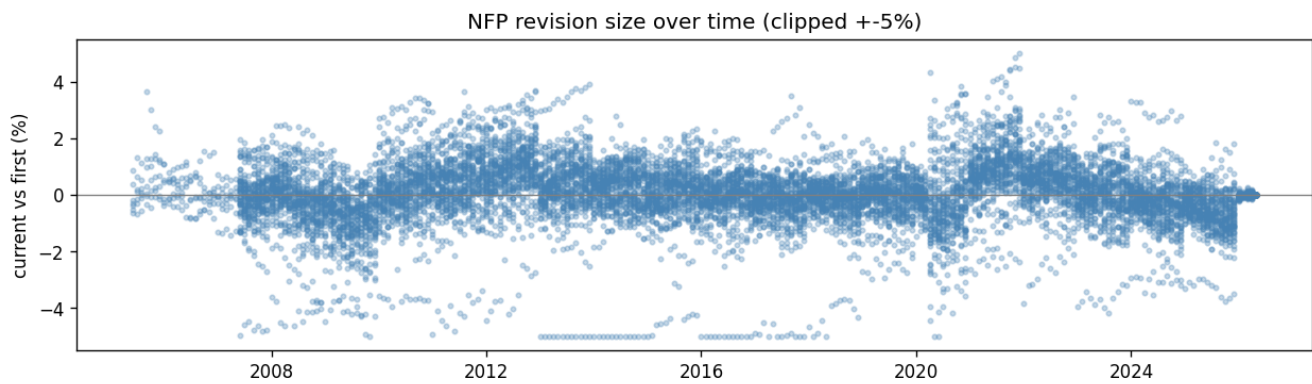
nfp_fig_10. State NFP release minus national jobs-report date



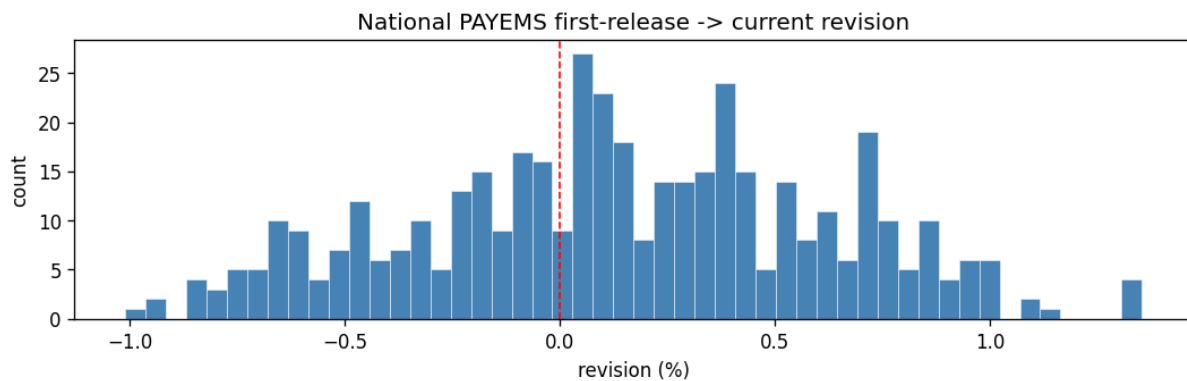
nfp_fig_11. State NFP first-release capture weekday



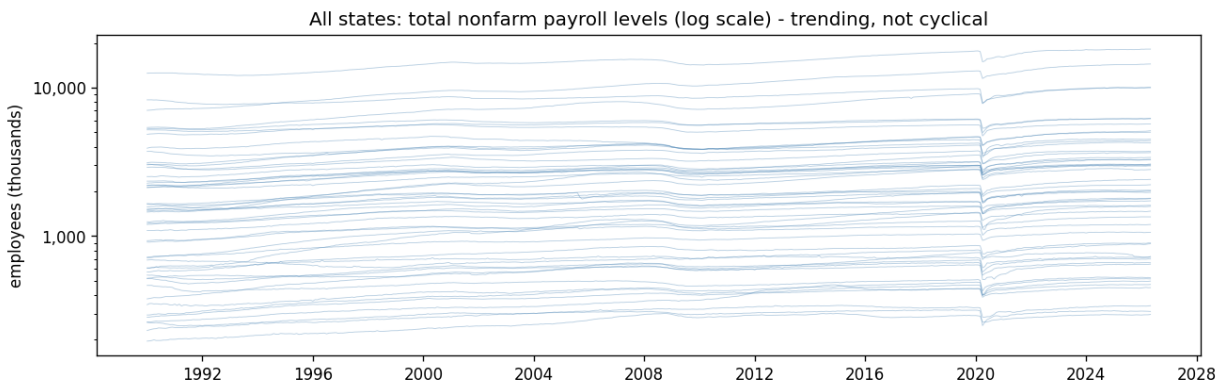
nfp_fig_12. Mean absolute revision by state (top 15)



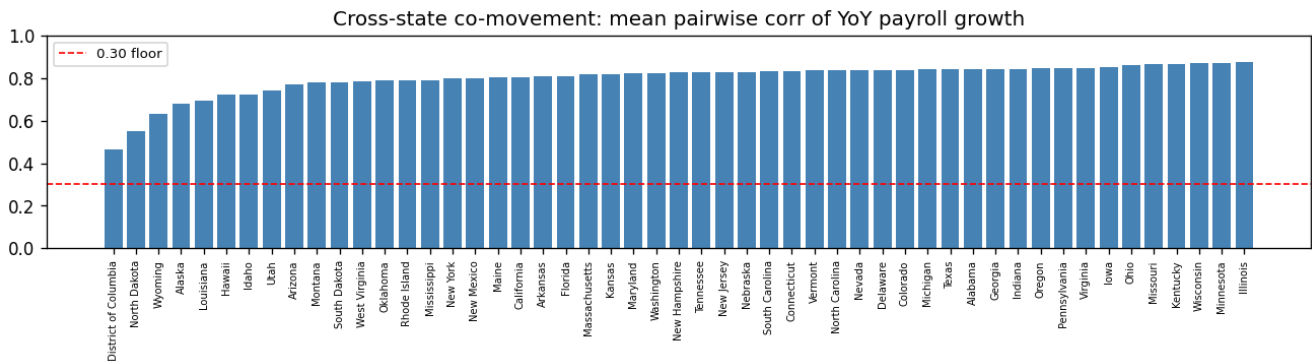
nfp_fig_13. NFP revision size over time (clipped +-5%)



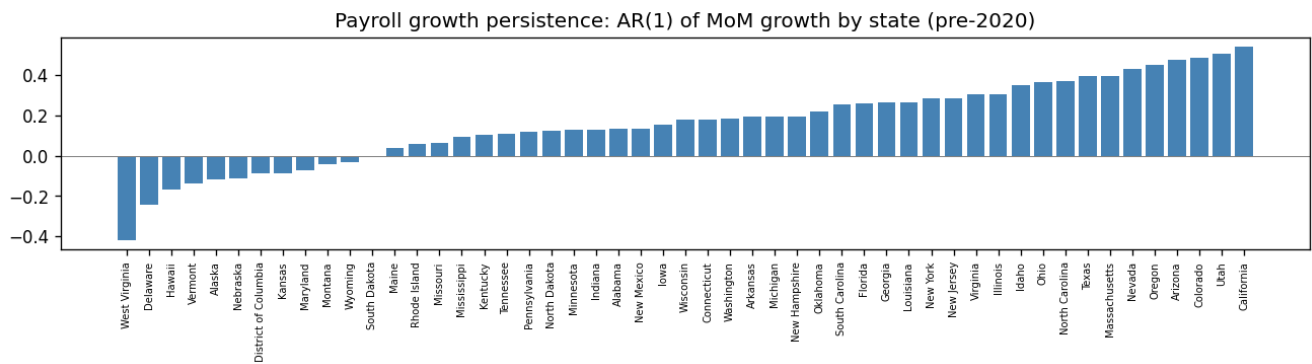
nfp_fig_14. National PAYEMS first-release -> current revision



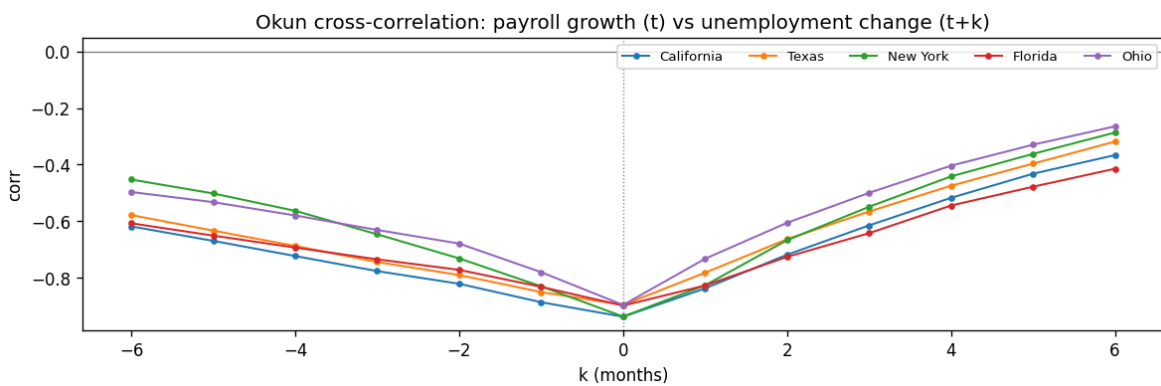
nfp_fig_15. All states: total nonfarm payroll levels (log scale) - trending, not cyclical



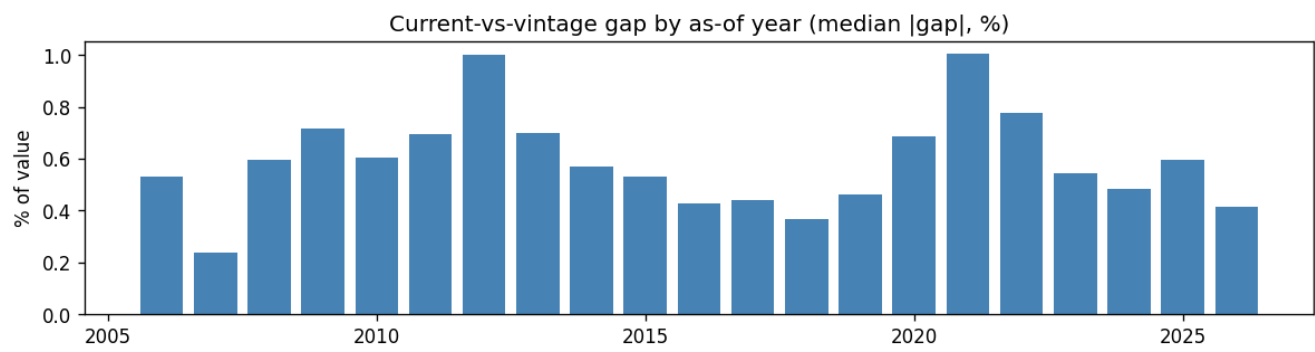
nfp_fig_16. Cross-state co-movement: mean pairwise corr of YoY payroll growth



nfp_fig_17. Payroll growth persistence: AR(1) of MoM growth by state (pre-2020)



nfp_fig_18. Okun cross-correlation: payroll growth (t) vs unemployment change (t+k)



nfp_fig_19. Current-vs-vintage gap by as-of year (median |gap|, %)

Appendix A - Full check results (all three datasets)

Every check with the question it answers, the result, and the measured value. PASS = criterion met; WARN = usable with the documented rule; FAIL = blocker.

Continued claims ({ABBR}CCLAIMS (weekly, NSA)) - 23 PASS / 20 WARN / 0 FAIL

ID	Question	Res	Metric
V1.1	Do we have all 51 states in every table?	PASS	{'vintage': 51, 'current': 51, 'pit': 51}
V1.2	Is (state, claim_type, week) a unique key?	PASS	vintage=0, current=0, pit=0
V1.3	Are key fields complete (no nulls)?	PASS	0 nulls
V1.4	Is every value parseable, finite and positive?	WARN	vintage bad=0; current zeros=15 (latest 1986-07-19), post-1990=0
V1.4b	Are zeros ramp-in artifacts (leading) or mid-sample dropouts (interior)?	WARN	9 leading, 6 interior; interior post-1990: 0
V1.5	Did any state's fetch silently truncate?	PASS	min=2099, max=2111
V2.1	Do obs dates follow the DOL Saturday week-ending convention?	PASS	0 non-Saturday
V2.2a	Are there missing weeks in the pre-1990 history?	WARN	48 missing weeks across 22 states (1986-88)
V2.2b	Are there missing weeks in modern data (live issues)?	WARN	Virginia 2026-05-23
V2.3	Do all states cover the same window?	PASS	start spread=5w, end spread=0w
V3.1	How far back can point-in-time be reconstructed?	WARN	first vintage 2010-09-16
V3.2	Does the vintage calendar match the DOL mid-week release schedule?	PASS	Thu 96.7%, Wed 3.3% of 867 vintages
V3.3	Is the national first release the advance print at obs+12d?	PASS	median lag 12d (min 11, max 61)
V3.4	On which weekday does FRED capture state data?	PASS	Mon 4.9%, Tue 0.0%, Wed 1.3%, Thu 60.1%, Fri 33.6%
V3.5	How long after the public release do state values appear?	PASS	32.6% within 0-2d of national release; 61.2% at +6-8d (next report)
V3.5b	Is the state-vs-national gap pattern stable within each era?	WARN	pre-2020: 97.8% at +6-8d; post-2020: 83.6% at +0-2d (+12.7% slip to +3-5d = Monday ingestion)
V3.6	Could a backtest have seen state data before it was public?	PASS	min state pub lag 13d across all 42k vintage rows
V3.7	Has FRED's capture timing / stored print type been stable?	WARN	switch ~2020-07-10: median 19d -> 13d
V3.8	Do weeks become available in order?	PASS	0 inversions
V3.9	Does the PIT panel contain any look-ahead?	PASS	0 violations, 0 unmatched keys of 41,667 rows
V4.1	How often does a first print get revised at all?	WARN	1402 of 42,074 obs revised (3.33%)
V4.1b	Where are revisions concentrated?	WARN	top-4 states = 56% of exceptions: Hawaii n=403 (med [0.2%]), Pennsylvania n=168 (med [0.1%]), New Mexico n=111 (med [3.0%]), Maryland n=104 (med [0.1%])
V4.2	Do values change mid-life and net out (invisible to endpoint checks)?	PASS	core states: 49 multi-valued obs of 10,556 (0.46%); Hawaii: 405 of 2152 (up to 5 values per obs)
V4.3	How big is the national advance-to-revised gap?	PASS	median -0.09%, IQR [-0.39%, +0.21%], max [6.4%]
V5.1	Are there frozen/stale stretches masquerading as data?	WARN	75 runs in current values; 34 in first-release values
V5.2	Which extreme moves are real events vs calendar mechanics vs unexplained?	WARN	1477 extreme moves: 224 COVID, 732 holiday, 167 pre-1990 era, 354 unexplained modern
V5.3	Do some states report rounded numbers (noise floor)?	WARN	1 states > 5%: Ohio 6%
V5.4	Did the TX 2024-06 +100k first-print error really happen, and which print was wrong?	WARN	first=48,748, current=148,748, neighbors mean=148,270
V6.1	Do the 51 states add up to the national total?	PASS	median ratio 0.986, IQR width 0.007, range [0.957, 1.188]
V6.2	Does every state co-move with the national cycle?	PASS	panel median 0.89; no state < 0.30
V7.1	How strong is the annual seasonality to handle?	WARN	median 0.64, min 0.17, max 0.85
V7.2	Are holiday weeks mechanically distorted?	WARN	holiday +0.100 vs regular -0.010 (spread +0.110 log-pts)
V7.3	Which transform makes the series stationary?	PASS	California: lvi p=0.00, yoy p=0.000; Texas: lvi p=0.00, yoy p=0.000; New York: lvi p=0.00, yoy p=0.000; Florida: lvi p=0.07, yoy p=0.000; Michigan: lvi p=0.00, yoy p=0.000
V7.4	Is persistence plausible in every state?	PASS	min 0.938 (New York), median 0.980

ID	Question	Res	Metric
V7.5	What did continued claims exclude during COVID?	WARN	2020-2021 levels exclude PUA/PEUC
V8.1	Are features indexed by information time?	PASS	all Fridays: True; median cclaims staleness 20d
V8.2	Do claims lead unemployment with the right sign (wiring check)?	PASS	California: peak corr 0.89 at k=0m; Texas: peak corr 0.88 at k=0m; New York: peak corr 0.90 at k=0m; Florida: peak corr 0.87 at k=0m; Michigan: peak corr 0.83 at k=0m
V8.3	How must weekly claims map onto monthly reference periods?	WARN	LAUS reference week = week containing the 12th
V9.1	From which date is the PIT panel backtest-complete?	PASS	51/51 from 2010-11-05; 0 holes after that
V9.2	Is the universe stable (no entry/exit)?	WARN	weeks per state: 2099-2106
V9.3	What staleness must a backtest assume, per era?	WARN	{'post-switch': {'median': 13.0, 'min': 13, 'max': 27}, 'pre-switch': {'median': 20.0, 'min': 13, 'max': 27}}
V9.4	How many feature rows carry later-corrected values?	WARN	t0 revised-later: 1386 rows (3.3%); any of t0..t7: 2823 rows (6.8%)
V10.1	Can the dataset be reproduced and audited byte-for-byte?	PASS	claims_validation_manifest.json

U3 unemployment ({ABBR}UR (monthly, SA rate)) - 26 PASS / 8 WARN / 0 FAIL

ID	Question	Res	Metric
V1.1	Do we have all 51 states in every table?	PASS	{'vintage': 51, 'UR current': 51, 'pit': 51}
V1.2	Is (state, series, month) a unique key?	PASS	vintage=0, current=0, pit=0
V1.3	Are key fields complete (no nulls)?	PASS	0 nulls
V1.4	Are all rates in a plausible bounded range?	PASS	0 out-of-range; observed [1.7, 30.5]%
V1.5	Are rates quantized to 0.1pp as published?	PASS	all values 1-decimal: True
V2.1	Are all observation dates month-starts?	PASS	0 non-first-of-month
V2.2	Are there missing months in any state?	WARN	1 month(s) missing across all states: 2025-10
V2.3	Do all states cover the same window?	PASS	start 1976-01-01..1976-01-01, end 2026-05-01..2026-05-01
V3.1	How far back can point-in-time be reconstructed?	WARN	first vintage 2005-07-22
V3.2	Does the vintage calendar match the jobs-report schedule?	PASS	Mon 2%, Tue 3%, Wed 4%, Thu 7%, Fri 84%
V3.3	Is the state publication lag ~50 days?	PASS	median 50d (IQR 48-53)
V3.4	Do state releases follow the national jobs report?	PASS	100.0% of state months released after national; median gap +14d
V3.5	Is the publication-lag regime stable over time?	WARN	median lag by year spans 48-66d (range 18d)
V3.6	Do months become available in order?	PASS	0 inversions
V3.7	Does the PIT panel contain any look-ahead?	PASS	0 violations, 0 unmatched of 11,133 rows
V4.1	How often/how much does a first print get revised?	WARN	86% of months revised; exact-zero 14%, within +-0.1pp 25%, within +-0.5pp 84%, max 4.1 pp
V4.2	Do larger revisions cluster at the benchmark?	PASS	99% of largest revisions land in Jan-Apr; modal release month = 3 (1581 of 1656 obs across CA/TX/NY)
V4.3	Is there a systematic bias in first prints?	WARN	mean revision -0.047pp (first prints overstate on average)
V4.4	How large is the national revision (calibration)?	PASS	median +0.00pp, IQR [+0.00, +0.10]pp, max 0.4 pp
V5.1	Which extreme moves are real (COVID) vs unexplained?	PASS	154 extreme moves: 140 COVID, 14 other
V5.2	Are there stale/frozen stretches?	WARN	42 runs of >= 12 identical months
V5.3	Is the COVID spike present and correctly sized?	PASS	Apr-May 2020 peak: median 13.5%, max 30.5% (Nevada)
V6.1	Does the LF-weighted mean track national UNRATE?	PASS	median weighted - national = 0.06pp vs unweighted 0.32pp
V6.2	Does every state co-move with the national cycle?	PASS	panel median 0.80; none < 0.30
V7.1	Did SA leave the rate free of residual seasonality?	PASS	month-of-year R2 of MoM pp change (SA): 0.004
V7.2	Is the rate persistent but bounded (not trending)?	PASS	California: AR1(level) 0.98, ADF p 0.02; Texas: AR1(level) 0.97, ADF p 0.02; New York: AR1(level) 0.96, ADF p 0.01; Florida: AR1(level) 0.98, ADF p 0.02; Ohio: AR1(level) 0.97, ADF p 0.08
V7.3	Are monthly pp changes stationary?	PASS	all sample changes ADF-stationary: True; median AR1(change) 0.75
V8.1	Are features indexed by information time?	PASS	monthly as-of grid: True; median data-staleness 2 months
V8.2	Does the rate move opposite to payrolls (Okun)?	PASS	California: corr -0.94; Texas: corr -0.90; New York: corr -0.94; Florida: corr -0.90; Ohio: corr -0.90
V8.3	How to align with monthly indicators?	WARN	UR references the week incl. the 12th; released with state NFP
V9.1	From which date is the PIT panel complete?	PASS	51/51 from 2008-08-01; 0 holes after
V9.2	Is the universe stable?	PASS	months per state: 604-604

ID	Question	Res	Metric
V9.3	How badly would current-instead-of-vintage contaminate a backtest?	WARN	median gap 0.20pp, p90 0.70pp, max 4.1pp
V10.1	Can the dataset be reproduced and audited?	PASS	ur_validation_manifest.json

Nonfarm payrolls ({ABBR}NA (monthly, SA level)) - 26 PASS / 8 WARN / 0 FAIL

ID	Question	Res	Metric
V1.1	Do we have all 51 states in every table?	PASS	{'vintage': 51, 'SA current': 51, 'pit': 51}
V1.2	Is (state, adjustment, month) a unique key?	PASS	vintage=0, current=0, pit=0
V1.3	Are key fields complete (no nulls)?	PASS	0 nulls
V1.4	Is every payroll level strictly positive and finite?	PASS	vintage bad=0, current bad=0
V1.5	Are the series lengths consistent across states?	PASS	min=437, max=437 months
V2.1	Are all observation dates month-starts?	PASS	0 non-first-of-month
V2.2	Are there missing months in any state?	PASS	0 gaps
V2.3	Do all states cover the same window?	PASS	start 1990-01-01..1990-01-01, end 2026-05-01..2026-05-01
V3.1	How far back can point-in-time be reconstructed?	WARN	first vintage 2005-07-22
V3.2	Does the vintage calendar match the jobs-report schedule?	WARN	Mon 3%, Tue 5%, Wed 5%, Thu 7%, Fri 80%
V3.3	Is the state publication lag ~50 days?	PASS	median 50d (IQR 48-53)
V3.4	Do state releases follow the national jobs report?	PASS	100.0% of state months released after national; median gap +14d
V3.5	Is the publication-lag regime stable over time?	WARN	median lag by year spans 48-66d (range 18d)
V3.6	Do months become available in order?	PASS	0 inversions
V3.7	Does the PIT panel contain any look-ahead?	PASS	0 violations, 0 unmatched of 11,133 rows
V4.1	How often/how much does a first print get revised?	WARN	99.2% of months revised; median rev 0.55%, p90 1.63%, max 7.5%
V4.3	Is there a systematic bias in first prints?	PASS	mean revision +0.052% (first prints understate on average)
V4.2	Do revisions cluster at the annual benchmark?	PASS	95% of largest revisions land in Jan-Mar; modal release month = 3 (1249 of 1308 obs across CA/TX/NY)
V4.4	How large is the national benchmark (calibration)?	PASS	median +0.13%, IQR [-0.19%, +0.46%], max [1.4%]
V5.1	Which extreme moves are real (COVID) vs unexplained?	WARN	242 extreme moves: 167 COVID, 75 other
V5.2	Are there stale/frozen stretches?	WARN	8 runs
V5.3	Are there level breaks beyond COVID?	PASS	21 state-months with MoM dlog > 2% outside 2020
V6.1	Do the 51 states sum to national PAYEMS?	PASS	median ratio 1.0001, range [0.9931, 1.0042]
V6.2	Does every state co-move with the national cycle?	PASS	panel median 0.83; none < 0.30
V7.1	Did SA actually remove the seasonality?	PASS	month-of-year R2 of MoM growth: SA 0.01 vs NSA 0.88
V7.2	Are levels trending and growth stationary?	PASS	California: lvl p=0.90, growth p=0.000; Texas: lvl p=0.78, growth p=0.000; New York: lvl p=0.74, growth p=0.000; Florida: lvl p=0.84, growth p=0.000; Ohio: lvl p=0.32, growth p=0.000
V7.3	Is payroll growth persistent?	PASS	median AR1 0.18, range [-0.42, 0.54]
V8.1	Are features indexed by information time?	PASS	monthly as-of grid: True; median data-staleness 2 months
V8.2	Does payroll move opposite to unemployment (Okun)?	PASS	California: corr -0.94; Texas: corr -0.90; New York: corr -0.94; Florida: corr -0.90; Ohio: corr -0.90
V8.3	How to align with monthly indicators?	WARN	both NFP and LAUS reference the week/pay-period incl. the 12th
V9.1	From which date is the PIT panel complete?	PASS	51/51 from 2008-08-01; 0 holes after
V9.2	Is the universe stable?	PASS	months per state: 437-437
V9.3	How badly would current-instead-of-vintage contaminate a backtest?	WARN	median gap 0.59%, p90 1.68%, max 7.5% between current and real-time latest value
V10.1	Can the dataset be reproduced and audited?	PASS	nfp_validation_manifest.json

Appendix B - Reproducibility

Each suite is self-contained: its notebook fetches every input from the FRED/ALFRED API and rebuilds its dataset from scratch. ALFRED queries with pinned realtime parameters are reproducible - a re-fetch reproduces byte-identical CSVs (SHA-256 verified).

Continued claims

Source: FRED/ALFRED api.stlouisfed.org — series {ABBR}CCLAIMS, {ABBR}ICCLAIMS, CCNSA

Vintage window: 2010-09-16 to 2026-06-26; current history 1986-01-04 to 2026-06-20.

File	Rows	SHA-256
claims_vintage_first_release.csv	84,148	8652317b8e781e8283c6670c3349c888f0ee4895061cea9391a942e18b6807d4
claims_current_history.csv	214,807	c31aec729507ec700ec49f71f8cee95a73a50dc614b36548d848d312a90ca582
claims_pit_features.csv	41,667	4d7abb21a38e9c8ac357de3fc96b3a2d3681775491ff685a003f21cfb25d574a

U3 unemployment

Source: FRED/ALFRED - series {ABBR}UR (SA U3 rate), {ABBR}LF (labor force), UNRATE

Vintage window: 2005-07-22 to 2026-06-23; current history 1976-01-01 to 2026-05-01.

File	Rows	SHA-256
ur_vintage_first_release.csv	11,745	20b3df686766ac7e7000bcb30119600ce22f8e06a8aba6bb126a648d1e9e5889
ur_current_history.csv	61,608	365865dfa20ca64ef6214ad780fdd4ea6de1ef530ef8e7e98f3ad2d716daf19f
ur_pit_features.csv	11,133	73af411d72dfcc5e8ff9f1d854a29373f5d9c0ae7d4c476c22b06a09470b24ad

Nonfarm payrolls

Source: FRED/ALFRED - series {ABBR}NA (SA total nonfarm), {ABBR}NAN (NSA), PAYEMS

Vintage window: 2005-07-22 to 2026-06-23; current history 1990-01-01 to 2026-05-01.

File	Rows	SHA-256
nfp_vintage_first_release.csv	11,796	878dea064d00badb4744ac2be2d0ad7fb3ffb08e20c010ba1ff6f9ccd51e0681
nfp_current_history.csv	44,574	c9b51e8c160ad56dba9ce943e1c407e71306d4bf302fc144127e5baa9259fe6f
nfp_pit_features.csv	11,133	95cd18dc427fe3dcccdf21850e15c319319dbba7e6446dff7f9be046854d039a1